PHYSICS

1)

For any spherical mirror focal length will be :-

(1) maximum for red light
(2) maximum for blue light
(3) maximum for white light
(4) Same for light of all colours

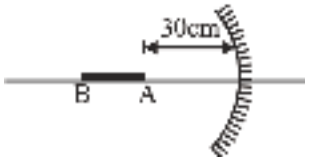
2) An object of height 7.5 cm is placed in front of a convex mirror of radius of curvature 25 cm at a distance of 40 cm. The height of the image should be -

(1) 2.3 cm
(2) 1.78 cm
(3) 1 cm
(4) 0.8 cm

3) A square loop of side 1 cm is placed perpendicular to the principal axis of a concave mirror of focal length 15cm at a distance of 20 cm. The area enclosed by the image of the wire is :-

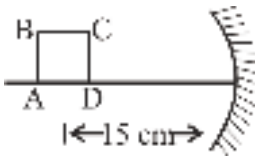
(1) 4 cm
(2) 6 cm^2
(3) 2 cm^2
(4) 9 cm^2

4) An object AB is placed on the axis of a concave mirror of focal length 10 cm. End A of the object is at 30 cm from the mirror. Find the length of the image. If length of object is 5 cm.



(1) 1 cm
(2) 2 cm
(3) 3 cm
(4) 4 cm

5) A squre ABCD of side 1 mm is kept at dist. 15cm in front of concave mirror as shown in figure. The focal length of mirror is 10 cm. The length of the perimeter of it's image will be :-



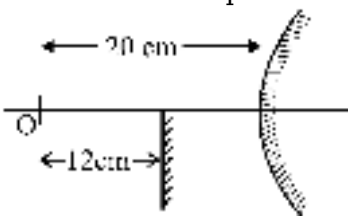
(1) 8 mm

(2) 2 mm

(3) 12 mm

(4) 6 mm

6) A luminous object is placed at a distance of 20 cm from convex mirror. A plane mirror is set so that virtual image is formed by both mirror will coincide. If plane mirror is at distance of 12 cm from object find out the focal length of mirror :-



(1) +5 cm

(2) –5 cm

(3) 15 cm

(4) 25 cm

7) A concave mirror is used to form an image of the sun on a white screen. If the lower half of the mirror is covered with an opaque card, the effect on the image on the screen would be -

(1) Negligible

(2) To make the image less bright than before

(3) To make the upper half of the image disappear

(4) To make the lower half of the image disappear

8) A man runs towards a plane mirror at a speed 15 m/s The speed of the image relative to the man is :-

(1) 15 ms^{-1}

(2) 30 ms^{-1}

(3) 35 ms^{-1}

(4) 20 ms^{-1}

9) An object is 20 cm away from a concave mirror of focal length 15 cm. If the object moves with a speed of 5 cm/s along principle axis, then the speed of the image will be :-

(1) 45 cm/s

(2) 27 cm/s

(3) 9 cm/s

(4) 10 cm/s

10) Distance of Real object & its real image from the focus of concave mirror are 16 cm & 9 cm respectively, then find focal length of mirror

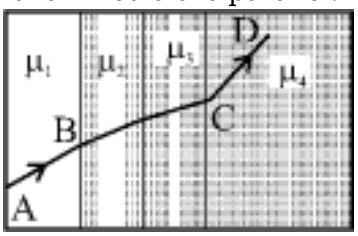
(1) 20 cm

(2) 15 cm

(3) 12 cm

(4) 8 cm

11) A ray of light passes through four transparent media with refraction indices μ_1, μ_2, μ_3 and μ_4 as shown in the given figure.
The surfaces of all media are parallel. If the emergent ray CD is parallel to the incident ray AB, we must have :-



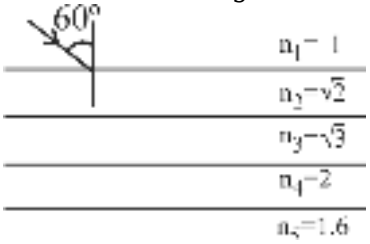
(1) $\mu_1 = \mu_2$

(2) $\mu_2 = \mu_3$

(3) $\mu_3 = \mu_4$

(4) $\mu_4 = \mu_1$

12) In the figure shown the angle made by the light ray with the normal in the medium of refractive index $\sqrt{3}$ is :



(1) 30°

(2) 60°

(3) 90°

(4) None of these

13) If refractive index of glass w.r.t. oil is $\frac{3}{5}$. Find refractive index of water. w.r.t oil is $\left[\mu_g = \frac{3}{2}, \mu_w = \frac{4}{3}\right]$

(1) $\frac{15}{8}$

(2) $\frac{6}{20}$

(3) $\frac{8}{15}$

(4) $\frac{20}{6}$

14) An air bubble is situated in glass slab of length 25 cm. When it is seen from one side, it appears at 6 cm and when see from other side it appears at 4 cm. Find refractive index of slab :-

(1) 1.5
(2) 2.5
(3) 1
(4) 1.33

15) A vessel is quarter filled with a liquid of refractive index μ. The remaining parts of the vessel is filled with an immiscible liquid of refractive index $\frac{3}{2}\mu$. The apparent depth of the vessel is 50% of the actual depth. The value of μ is :-

(1) 1
(2) $\frac{3}{2}$
(3) $\frac{2}{3}$
(4) $\frac{4}{3}$

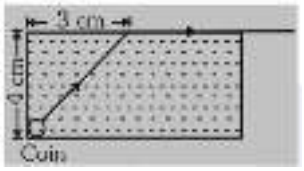
16) The apparent depth of a needle lying at the bottom of the tank, which is filled with water of refractive index 1.33 to a height of 12.5 cm is measured by a microscope to be 9.4 cm. If water is replaced by a liquid of refractive index 1.5 upto the same height. What distance would the microscope have to be moved to focus on the needle again ?

(1) 1.1 cm
(2) 2.13 cm
(3) 3.1 cm
(4) 4.15 cm

17) Light travels in two media A & B with speeds 1.8×10^8 m/s & 2.4×10^8 m/s respectively. Then the critical angle between them is :-

(1) $\sin^{-1}(2/3)$
(2) $\tan^{-1}(3/4)$
(3) $\tan^{-1}(2/3)$
(4) $\sin^{-1}(3/4)$

18) A small coin is resting on the bottom of a beaker filled with a liquid. A ray of light from the coin travels upto the surface of the liquid and moves along the surface (see figure).



How fast is the light travelling in the liquid ?

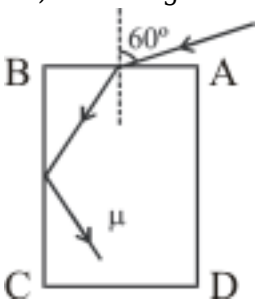
(1) 1.2×10^8 m/s

(2) 2.4×10^8 m/s

(3) 1.8×10^8 m/s

(4) 3.0×10^8 m/s

19) In the figure the minimum value of refractive index if T.I.R. occurs on surface BC :-



(1) 1.732

(2) 1.52

(3) 1.32

(4) 1.12

20) A ray of light from a denser medium strike a rarer medium. The angle of reflection is r and that of refraction is r'. The reflected ray & refracted ray makes an angle of 90° with each other. The critical angle will be :-

(1) $\sin^{-1}$ (tan r)

(2) $\tan^{-1}$ (sin r)

(3) $\sin^{-1}$ (tan r')

(4) $\tan^{-1}$ (sin r')

21) A thin rod of length $f/3$ lies along the axis of a concave mirror of focal length f. One end of its magnified image touches an end of the rod. The length of the image is :-

(1) f

(2) $\frac{1}{2f}$

(3) $2f$

(4) $\frac{1}{4f}$

22) When an object is placed at a distance of 25 cm from a concave mirror, the magnification is m_1. The object is moved 15 cm farther away with respect to the earlier position, and the magnification becomes m_2. If $m_1/m_2 = 4$ the focal length of the mirror is (Assume image is real m_1, m_2 are

numerical values)

(1) 10 cm

(2) 30 cm

(3) 15 cm

(4) 20 cm

23) A variable force ($24t^2$ – 8t) Newton is acting on a particle. The impulse imparted to it in first two seconds will be :

(1) 12 kg-m/s

(2) 24 kg-m/s

(3) 36 kg-m/s

(4) 48 kg-m/s

24) An iron ball of mass m = 50 gm falls from a hight of 5 m and rised upto 3.2 m after colliding with the horizontal surface. If the time of contact of the ball and surface is 0.02 s. Find magnitude average contact force exerted on the ball by the horizontal surface :-

(1) 55 N

(2) 45 N

(3) 90 N

(4) 22 N

25) A particle of mass 2kg moves in free space such that its position vector varies with time as $\vec{r} = \left[\left(3+4t^2\right)\hat{i} + (2t)\hat{j} + (3-6t)\hat{k}\right]$ m where t is in seconds. Net force acting on the particle is

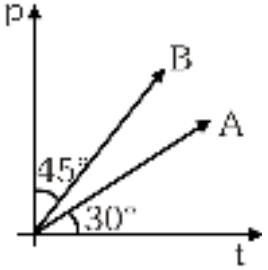
(1) zero

(2) parallel to x-axis

(3) parallel to y-axis

(4) time dependent

26) For the given momentum-time (p-t) graphs find the ratio of force on particles A & B.



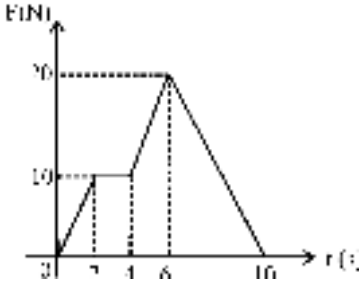
(1) $\sqrt{3}$

(2) $\frac{1}{\sqrt{3}}$

(3) 1

(4) $\frac{3}{4}$

27) A particle of mass 2 kg is initially at rest. A force acts on it whose magnitude changes with time. The force time graph is shown below :-



The velocity of the particle after 10 s is

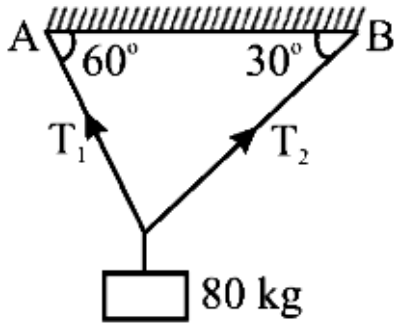
(1) 20 ms^{-1}

(2) 50 ms^{-1}

(3) 75 ms^{-1}

(4) 100 ms^{-1}

28) A block of mass 80 kg is supported by two cables as shown in the figure. Then the ratio of tensions T_1 and T_2 is :-



(1) 1 : 1

(2) $1 : \sqrt{3}$

(3) $\sqrt{3} : 1$

(4) 1:3

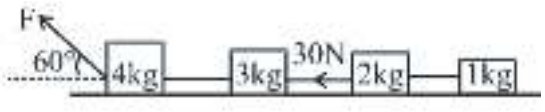
29) 

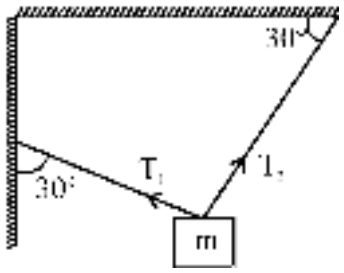
Figure shows four blocks that are being pulled along a smooth horizontal surface. The masses of the blocks and tension in one cord are given. The pulling force F is :-

(1) 50 N

(2) 100 N

(3) 125 N

(4) 200 N

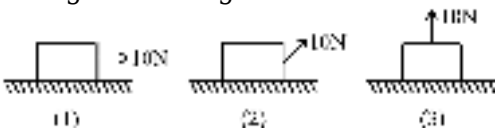
30) Calculate T_1 & T_2 :- 

(1) $\frac{\sqrt{3}}{2}mg, \frac{mg}{2}$

(2) $mg, \frac{\sqrt{3}}{2}mg$

(3) $\frac{\sqrt{3}}{2}mg, \frac{\sqrt{3}}{2}mg$

(4) $\frac{mg}{2}, \frac{mg}{2}$

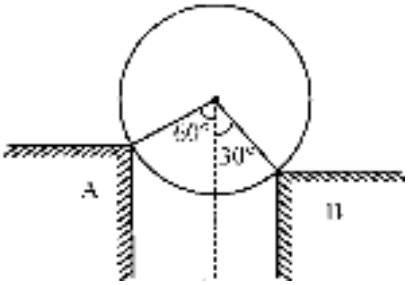
31) A block rests on a horizontal surface and a man pulls it with a 10N force. Rank the situation shown below according to the magnitude of the normal force exerted by the surface on the block, least to greatest. 

(1) 1, 2, 3

(2) 2, 1, 3

(3) 3, 2, 1

(4) 1, 3, 2

32) A smooth cylinder of mass m and radius R is resting on two corner edges A and B as shown in fig. The relation between normal reaction at the edges A and B is :- 

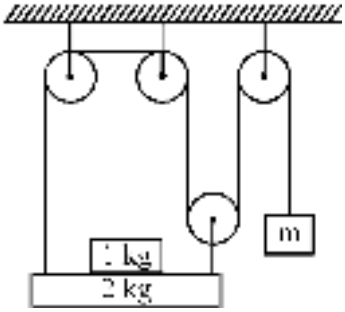
(1) $N_A = \sqrt{2}N_B$

(2) $N_B = \frac{2\sqrt{3}N_A}{5}$

(3) $N_A = \frac{N_B}{2}$

(4) $N_B = \sqrt{3}N_A$

33) If the arrangement shown in the figure is in equilibrium then mass m of the block will be :-

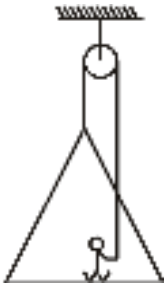


(1) 3 kg

(2) 2 kg

(3) 1.5 kg

(4) 1 kg

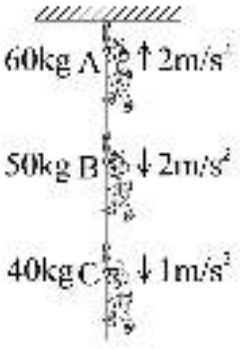
34) A man of mass 50 kg stands on a frame of mass 30kg. He pulls on a light rope which passes over a pulley. The other end of the rope is attached to the frame. For the system to be in equilibrium what force man must exert on the rope ? 

(1) 400 N

(2) 800 N

(3) 300 N

(4) 500 N



35) Tension in the rope at the rigid support is : ($g = 10\ m/s^2$)

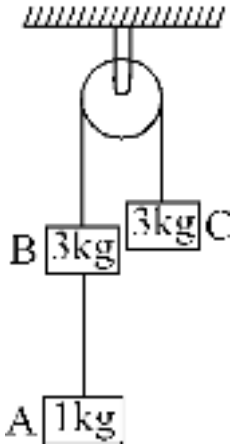
(1) 760 N

(2) 1360 N

(3) 1480 N

(4) 1620 N

36) In the adjoining figure, the tension in the string connecting between A and B is :-



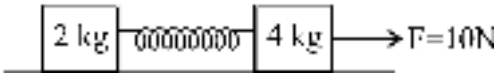
(1) $\frac{6}{7}g$

(2) $\frac{g}{7}$

(3) $\frac{8g}{7}$

(4) g

37) Two blocks of mass 2 kg and 4 kg are accelerated with same acceleration by a force 10 N as shown in figure on smooth horizontal surface. then the spring force between the two blocks will be :



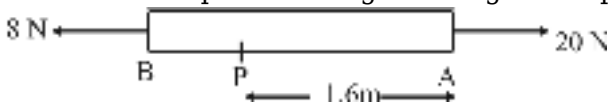
(1) 5 N

(2) 10 N

(3) $\frac{10}{3}$ N

(4) $\frac{5}{3}$ N

38) In the given diagram mass of rope AB is 4 kg and length of rope is 2 m. Calculate the tension at point P in the rope :-



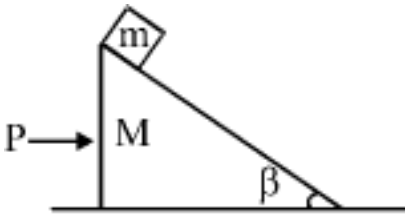
(1) 4.8 N

(2) 10.4 N

(3) 8.8 N

(4) 15.2 N

39) Two blocks are moving on a smooth horizontal surface such that the mass m remains stationary with respect to block of mass M as shown in the figure. The magnitude of force P is

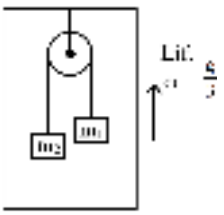


(1) (M + m)g tan β

(2) g tan β

(3) mg cos β

(4) (M + m)g cosec β



40) If $m_1 : m_2 = 2 : 1$, then acceleration of bodies with respect to lift is –

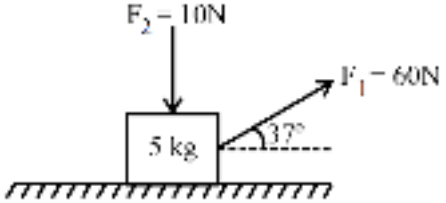
(1) $\frac{4g}{3}$

(2) $\frac{2g}{3}$

(3) $\frac{4g}{9}$

(4) $\frac{5g}{9}$

41) If (μ = 0.5) then find out acceleration of block :



(1) 25 m/s^2

(2) 12 m/s^2

(3) 7.2 m/s^2

(4) 0

42) A block of 10 kg is kept over a rough surface and a force F = 4t is applied on it. Then the minimum value of t for which the block will start moving, is (g = 10 m/s^2):-

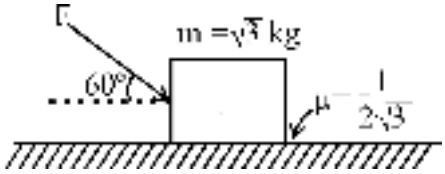


(1) 4 sec

(2) 10 sec

(3) 8 sec

(4) 5 sec

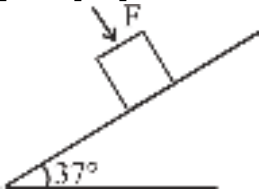
43) What is the maximum value of the force F such that the block shown in the arrangement does not move :- 

(1) 20 N

(2) 10 N

(3) 12 N

(4) 15 N

44) A block of mass 2 kg is resting on a rough inclined plane of inclination 37° as shown in figure. The coefficient of friction between the block and the plane is μ = 0.5. What minimum force F should be applied perpendicular to the plane on the block so that block does not slip on the plane. (g = 10 m/s^2) 

(1) Zero

(2) 6 N

(3) 8 N

(4) 10 N

45) A block takes twice as much time to slide down a 45° rough inclined plane as it takes to slide down a similar smooth plane. The coefficient of friction is :

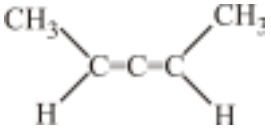
(1) $\frac{3}{4}$

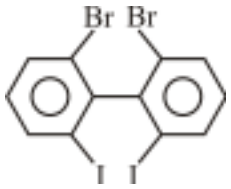
(2) $\frac{\sqrt{3}}{2}$

(3) $\frac{1}{4}$

(4) $\frac{1}{3}$

CHEMISTRY

1) **Statement-I :-** Penta-2,3-diene  is optically active.

Statement-II :- Given O,O'-di substituted Biphenyls 

is optically active due to perpendicular plane because of repulsion factor caused by bulky groups.
Identify correct option :-

(1) Both statement-I & statement-II are correct

(2) Both statement-I & statement-II are incorrect

(3) Statement-I is correct but statement-II is incorrect

(4) Statement-I is incorrect but statement-II is correct

2) Choose the correct option.
Assertion (A) : Staggered conformation of ethane is more stable than its eclipsed conformation.
Reason (R) : Of all the conformations of ethane the staggered form has the least torsional strain and the eclipsed form has the maximum torsional strain.

(1) Both 'A' and 'R' are correct and 'R' is the correct explanation of 'A'

(2) Both 'A' and 'R' are correct but 'R' is not correct explanation of 'A'

(3) 'A' is correct but 'R' is incorrect

(4) 'A' is incorrect but 'R' is correct

3) Total number of isomer's having molecular formula C_4H_8 :-

(1) 2

(2) 3

(3) 4

(4) 6

4) How many optically active stereoisomers are possible for Butane-2,3-diol ?

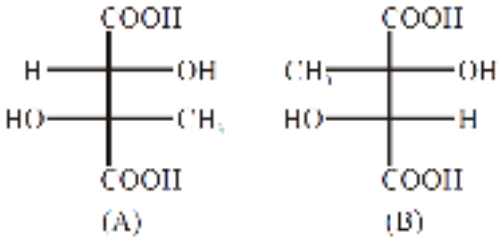
(1) 2

(2) 1

(3) 4

(4) 3

5) The structures A and B represent :-



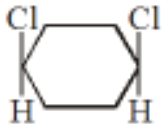
(1) enantiomers

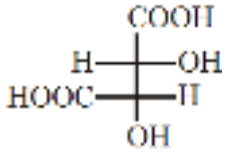
(2) diastereomers

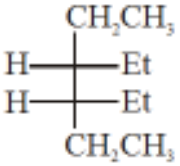
(3) homomers

(4) racemic mixture

6) Which of the following is a meso compound:-

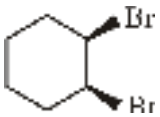
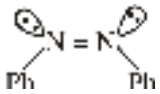
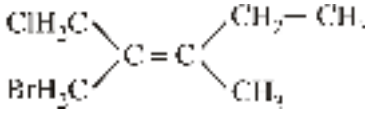
(1) 

(2) 

(3) 

(4) All of the above

7) Match List-I with List-II and select the answer from the given codes.

List-I Structure of the compound		**List-II Name of the geometrical isomerism**	
(i)		(a)	Cis
(ii)		(b)	E-isomer
(iii)		(c)	Syn
(iv)		(d)	Trans

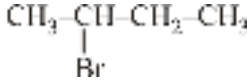
(1) i-a, ii-c, iii-b, iv-d

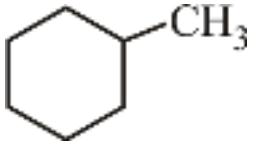
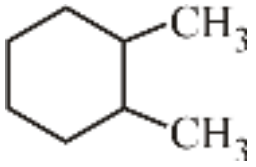
(2) i-a, ii-b, iii-c, iv-d

(3) i-b, ii-a, iii-c, iv-d

(4) i-c, ii-a, iii-d, iv-b

8) Match the column ?

Compound		**Chiral center**	
(P)		(i)	2

(Q)		(ii)	1
(R)		(iii)	0

(1) P-ii, Q-iii, R-i

(2) P-i, Q-ii, R-iii

(3) P-ii, Q-i, R-iii

(4) P-iii, Q-ii, R-i

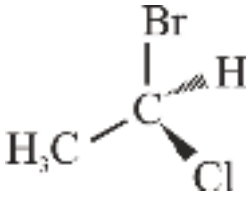
9) Select incorrect statement :

(1) Most stable conformation of cyclohexane is "Chair" form

(2) Generally in n-butane, Anti > Gauche > Eclipsed (Stability)

(3) To show geometrical isomerism, compound must have double bond

(4) Minimum number of carbon in a hydrocarbon to show chain isomerism is "four"

10) Configuration of compound is :- 

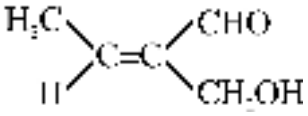
(1) E

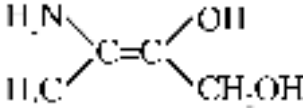
(2) R

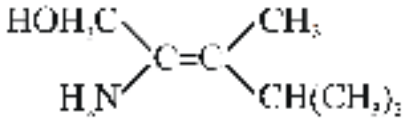
(3) S

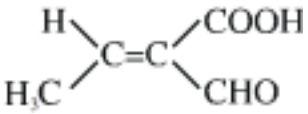
(4) Z

11) Which of the following has E-configuration :-

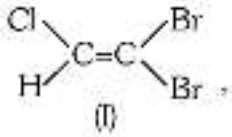
(1) 

(2) 

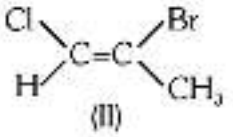
(3) 

(4) 

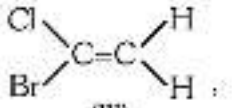
12) Which is a pair of geometrical isomers :-



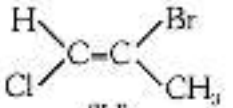
(I)



(II)



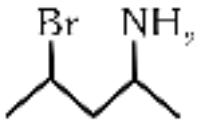
(III)

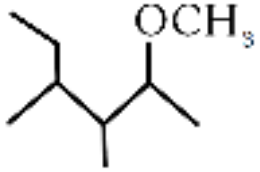


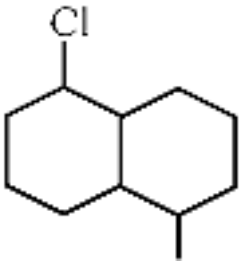
(IV)

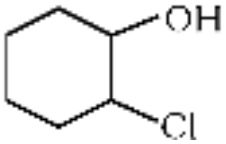
(1) I and II
(2) I and III
(3) II and IV
(4) III and IV

13) The compound which has maximum number of chiral carbon is :-

(1) 

(2) 

(3) 

(4) 

14)

Which conformer of the given compound is most stable (rotation about C–C bond) :-
$HO–CH_2–CH_2–NO_2$

(1) Staggered
(2) Gauche
(3) Fully eclipsed
(4) Partially eclipsed

15) Which of the following is not a pair of isomers :

(1) Propyne and Cyclopropene
(2) Propyne and Propadiene

(3) Propene and Cyclopropene

(4) 1-Propanol and Methoxy ethane

16) Which of the following shows geometrical isomerism :-
(a) 2–chloro but–2–ene
(b) 1–phenyl propene
(c) 1, 3–dichloro cyclopentene
(d) 2–methyl pent–2–ene

(1) a, b, c,

(2) a, c

(3) a, c, d

(4) a, b

17) How many primary amines are possible by the formula $C_4H_{11}N$:-

(1) 1

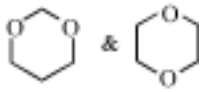
(2) 2

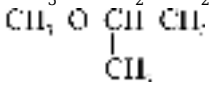
(3) 3

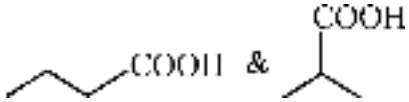
(4) 4

18) Which of following pairs of compounds are not position isomers ?

(1) 

(2) 

(3) $CH_3O–CH_2–CH_2–CH_3$ &


(4) 

19) How many position isomers are possible when three hydrogens of benzene are replaced by three chlorines :-

(1) 3

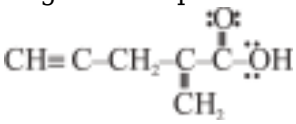
(2) 6

(3) 10

(4) 4

20)

In given compound



which is not correct ?

(1) D.U. = 4

(2) Acetylinic bond = 1

(3) π – bond = 4

(4) Olefinic bond = 2

21) How many structural isomer of C_6H_{12} having 4-member ring ?

(1) 3

(2) 4

(3) 5

(4) 6

22) Minimum number of carbon atoms that must be present in alkyne to show chain and position isomerism, respectively is :-

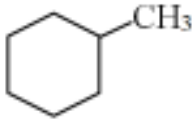
(1) 5,5

(2) 6,4

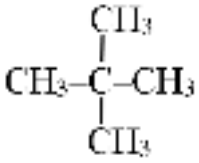
(3) 5,4

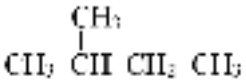
(4) 4,5

23) Which of the following form only one mono chloro product when halogenated ?

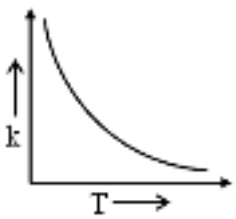
(1) 

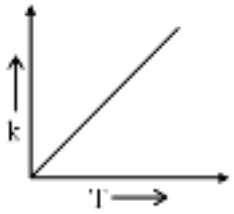
(2) $CH_3–CH_2–CH_2–CH_2–CH_3$

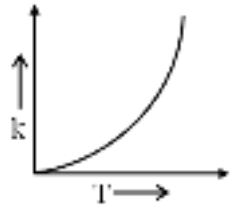
(3) 

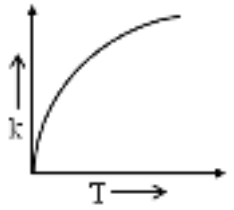
(4) 

24) Which one of the following given graphs represents the variation of rate constant (k) with temperature (T) for an endothermic reaction ?

(1) 

(2) 

(3) 

(4) 

25) For a 1st order reaction, slope of ln k vs $\frac{1}{T}$ graph is –4000 then activation energy in kcal/mol is :

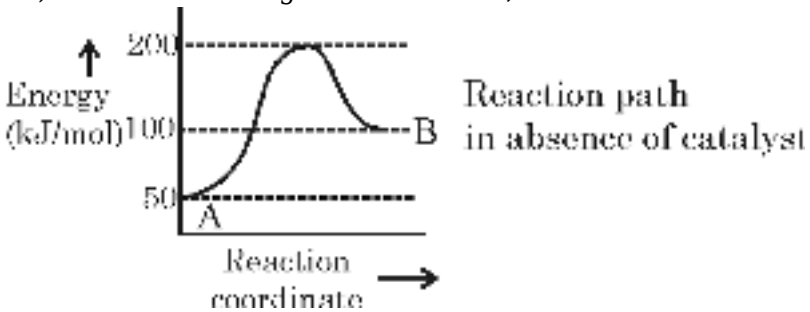
(1) 8000

(2) 8

(3) 33.2

(4) 800

26) For a reversible gaseous reaction, A ⇌ B



Which of the following is an incorrect statement ?

(1) ΔH of reaction is 50 kJ/mol

(2) Activation energy for backward reaction is 100 kJ/mol in absence of catalyst

(3) In presence of positive catalyst activation energy of forward reaction will become 130 kJ/mol, if catalyst reduces threshold energy to 180 kJ/mol

(4) In presence of positive catalyst, ΔH is less than 50 kJ/mol

27) Rate constant varies with temperature by the equation $\log_{10}k = 5 - \frac{2000}{T}$. We can conclude that ($R = 8.314\ Jmol^{-1}K^{-1}$) :-

(1) Pre exponential factor A is 5

(2) E_a is 4 kcal/mol

(3) Pre exponential factor A is 10^5

(4) E_a is 19.212 kcal/mol

28) Which of the following will affect the rate constant?

(1) Concentration of reactant

(2) Temperature

(3) Pressure

(4) Concentration of product

29) Which factors would increase the rate of a chemical reaction?
I. Increasing the temperature
II. Addition of reactant
III. Adding a catalyst

(1) I and II

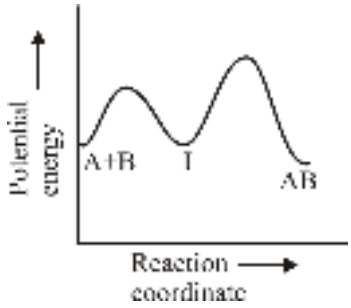
(2) II and III

(3) I, II and III

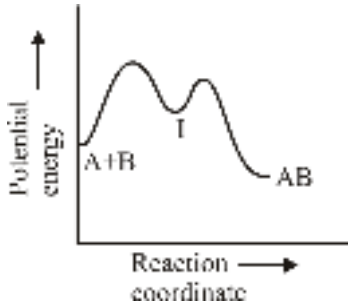
(4) I and III

30) For an exothermic reaction, following two steps are involved.
Step 1. $A + B \rightarrow I$ (slow)
Step 2. $I \rightarrow AB$ (fast) Which of the following graphs correctly represent this reaction?

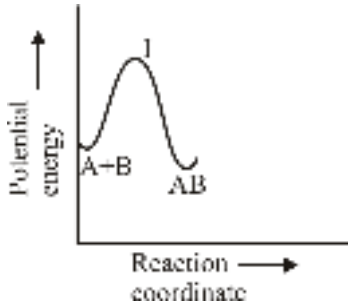
(1)

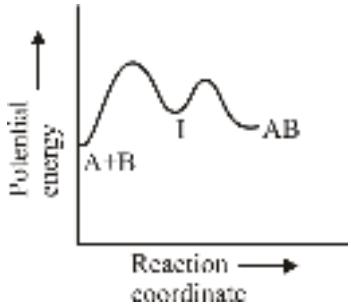


(2)



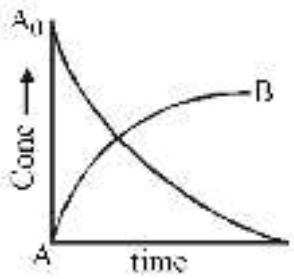
(3)



(4) 

31) For a complex reaction $A \xrightarrow{k}$ products
E_{a_1} = 180 kJ/mol ; E_{a_2} = 80 kJ/mol ; E_{a_3} = 50 kJ/mol Overall rate constant k is related to individual rate constant by the equation $k = \left(\frac{k_1 . k_2}{k_3}\right)^{2/3}$.
Activation energy (kJ/mol) for the overall reaction is :-

(1) 100
(2) 43.44
(3) 150
(4) 140

32) For the reaction, A → nB, At the point of intersection of the two curves shown, the conc. of B is given by :- 

(1) $\frac{nA_0}{2}$
(2) $\frac{A_0}{n-1}$
(3) $\frac{nA_0}{n+1}$
(4) $\left(\frac{n-1}{n+1}\right) A_0$

33) For a first order reaction rate constant is given as $\log k = 14 - \frac{1.2 \times 10^4}{T}$, then what will be the value of temperature if its half life period is 6.93×10^{-3} min

(1) 100 K
(2) 1000 K
(3) 720 K
(4) 327 K

34)

At 527°C, the reaction given below has $K_C = 4$

$$NH_3(g) \rightleftharpoons \frac{1}{2} N_2(g) + \frac{3}{2} H_2(g)$$

What is the K_p for the reaction ?

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

(1) $16 \times (800\ R)^2$

(2) $\left(\frac{800R}{4}\right)^{-2}$

(3) $\left(\frac{1}{4 \times 800R}\right)^2$

(4) None of these

35) $CH_4(g) + 2O_2(g) \rightleftharpoons CO_2(g) + 2H_2O(\square)$,

$\Delta_r H = -170.8\ kJ\ mol^{-1}$

Which of the following statements is not true?

(1) At equilibrium the concentrations of $CO_2(g)$ and $H_2O(\square)$ are not equal

(2) The equilibrium constant for the reaction is given by $K_p = \frac{[CO_2]}{[CH_4][O_2]}$

(3) Addition of $CH_4(g)$ or $O_2(g)$ at equilibrium will cause a shift to the right

(4) The reaction is exothermic

36) At which temperature the value of K_P becomes equal to K_C :-

(1) 273 K

(2) 1 K

(3) 12.19 K

(4) 546 K

37) For the reaction, $A(g) + 2B(g) \rightleftharpoons 2C(g) + D(s)$,
2 moles of A, 3 moles of B and 1 mole of C are present in 10L vessel. If K_c for the reaction is 3.6, the reaction :-

(1) Will proceed in Forward direction

(2) Will proceed in Backward direction

(3) Is at equilibrium

(4) None of these

38) In the reaction,
$A(s) + B(g) \rightleftharpoons 2C(s) + 2D(g)$ equilibrium is established. If the pressure of B is doubled then to re-establish the equilibrium, how many times of pressure of D becomes as compared to initial pressure-

(1) 2

(2) 3

(3) $\sqrt{2}$

(4) $\sqrt{3}$

39) Consider the following reversible reactions (at 298 K):-
(a) $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$
(b) $2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$
(c) $2HI_{(g)} \rightleftharpoons H_{2(g)} + I_{2(g)}$
(d) $X_{(g)} + Y_{(g)} \rightleftharpoons 4Z_{(g)}$

Highest and lowest value of $\frac{K_p}{K_c}$ will be shown by the equilibrium

(1) d, b
(2) a, c
(3) b, a
(4) b, c

40)

1 mole of PCl_5 is heated in a 1 litre closed vessel. At equilibrium, 20% moles of PCl_5 are not dissociated. What would be the value of K_C ?

(1) 2.4
(2) 42
(3) 0.32
(4) 3.2

41) 2 mole N_2 are mixed with 6 mole H_2 in a closed vessel of 1 L capacity. If 50%. N_2 is converted into NH_3 at equillibrium. The value of K_c for the reaction $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ is-

(1) $\frac{4}{27}$
(2) $\frac{27}{4}$
(3) $\frac{1}{27}$
(4) 24

42) In the reaction $3A_{2(g)} \rightleftharpoons 2A_{3(g)}$ equilibrium is obtained by taking initial concentration of A_2 as a mol/Litre then x mol amount is dissociated. The total moles at equilibrium will be :-

(1) a – aα
(2) $\frac{a\alpha}{3}$
(3) a – 3aα
(4) $a - \frac{a\alpha}{3}$

43) At a given temperature, equilibrium is attained when 50% of each reactant is converted into the products
$A(g) + B(g) \rightleftharpoons C(g) + D(g)$

If amount (in mol) of B(g) is doubled, percentage of B converted into products will be :-

(1) 100 %

(2) 50 %

(3) 66.67 %

(4) 33.33 %

44) For a gas reaction,
$3H_2(g) + N_2(g) \rightleftharpoons 2NH_3(g)$ at equillibrium, the partial pressure of H_2 and N_2 are 0.8 and 1.6 atm respectively and the total pressure of the entire system is 3.2 atm. What will be the value of K_p :-

(1) 0.58 atm^{-2}

(2) 0.78 atm^{-2}

(3) 0.88 atm^{-2}

(4) 0.48 atm^{-2}

45) Which of the following will not change the concentration of ammonia in the equilibrium :
$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$; $\Delta H = -x$ kJ

(1) increase of pressure

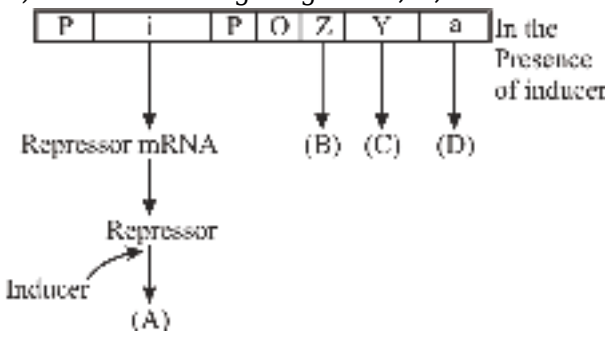
(2) increase of temperature

(3) decrease of volume

(4) addition of catalyst

BIOLOGY

1) In the following diagram A, B, C and D is respectively :-



(1) A– Inactive repressor, B– β galactosidase, C– Permease, D– Transacetylase

(2) A– Repressor, B– β galactosidase, C– Permease, D– Catalase

(3) A– Repressor, B– β galactosidase, C– Transacetylase, D– Permease

(4) A– Inactive Repressor, B– permease, C– Transacetylase, D– Catalase

2) Given below are two statements :
Statement-I : RNA mutates at a slower rate.

Statement-II : Viruses having RNA genome and short life span mutate and evolve faster.
In the light of the above statements, choose the correct answer from the options given below :

(1) Both statement-I and statement-II are false

(2) Statement-I is true but statement-II is false

(3) Statement-I is false but statement-II is true

(4) Both statement-I and statement-II are true

3) **Assertion :-** In Gel Electrophoresis, the DNA fragments resolve according to their size through the sieving effect provided by the agarose gel.
Reason :- The larger the DNA fragment size the farther it moves.
In the light of the above statements, choose the **correct answer** from the options given below :-

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Assertion is false but Reason is true.

4) Match the column-I with Column-II with resepect to mutation :-

Column-I		**Column-II**	
(A)	Frame-shift	(i)	Loss or gain of a segment of DNA
(B)	Chromosomal aberrations	(ii)	Replacement of Nitrogen base by another nitrogen base
(C)	Substitution	(iii)	Increase the set of chromosomes
(D)	Euploidy	(iv)	Deletions or insertions of nitrogen base pairs of DNA

(1) A-iv, B-i, C-iii, D-ii

(2) A-iv, B-i, C-ii, D-iii

(3) A-iv, B-i, C-iii, D-ii

(4) A-iv, B-iii, C-ii, D-i

5) **Assertion:** UTRs are present at both 5’ end and 3’end in mRNA.
Reason: UTRs are required for efficient translation process.

(1) Both the assertion and the reason are true and reason is a correct explanation of the assertion.

(2) Both the assertion and reason are true but reason is not a correct explanation of the assertion.

(3) Assertion is true but reason is false.

(4) Assertion and reason both are false.

6) Identify the incorrect match

(1) Ribozyme – Nucleic acid

(2) Spliceosome – Bacteria

(3) Genetic code – Triplet

(4) UGA - Stop codon

7) Match List-I with List-II :

	List-I		**List-II**
A.	Monocistronic gene	I.	Coding sequence
B.	Polycistronic gene	II.	Intervening sequence
C.	Exons	III.	Mostly prokaryotes
D.	Introns	IV.	Mostly eukaryotes

(1) (A–IV), (B–III), (C–I), (D–II)

(2) (A–III), (B–IV), (C–II), (D–I)

(3) (A–II), (B–I), (C–III), (D–IV)

(4) (A–I), (B–II), (C–IV), (D–III)

8) Match the column-I with Column-II

	Column-I		**Column-II**
A	hn RNA	i	RNA Polymerase
B	28 s rRNA	ii	RNA Polymerase II
C	Sn-RNA	iii	RNA Polymerase I
D	16 s rRNA	iv	RNA Polymerase III

	A	B	C	D
(1)	ii	iii	iv	i
(2)	iii	ii	i	iv
(3)	iii	ii	iv	i
(4)	ii	iii	i	iv

(1) 1

(2) 2

(3) 3

(4) 4

9) Choose the correct sequence w.r.t. translation process

(a) Ribosome moves from codon to codon.
(b) Initiator tRNA binds to codon in mRNA.
(c) Smaller subunit of ribosome binds to the mRNA at the start codon.
(d) Release factor binds to stop codon.

(1) a → c → d → b
(2) c → b → d → a
(3) c → b → a → d
(4) d → c → a → a

10) Who proposed that the genetic code for amino acids should be made up of three nucleotides ?

(1) George Gamow
(2) Francis Crick
(3) Jacque Monod
(4) Franklin Stahl

11) Which one of the following is a wrong statement regarding mutations ?

(1) Deletion and insertion of base pairs cause frame-shift mutations
(2) Cancer cells commonly show chromosomal aberrations
(3) UV and Gamma rays are mutagens
(4) Gain or loss of chromosome number in subset cause substitution mutation

12) Consider the following four statements (a-d) and select the option which include all the correct ones only:-
(a) Transcription and translation can be coupled in bacteria.
(b) The split gene arrangement represents the advanced feature of the genome.
(c) The process of splicing represents the dominance of RNA-world (d) The presence of introns in split gene is reminiscent of antiquity.

(1) a, b, and d
(2) a, b and c
(3) a, c, and d
(4) b, c, and d

13) When the codon of mRNA is 5' – GUC – 3', the anticodon on tRNA will be

(1) 5'–CAG–3'
(2) 3'–CAG–5'
(3) 3'–CUG–5'
(4) 3'–GAC–5'

14) Gel electrophoresis is used for -

(1) Construction of recombinant DNA by joining with cloning vectors.
(2) Isolation of DNA molecules from cell.

(3) Cutting of DNA into fragments.

(4) Separation of DNA fragments according to their size.

15)

Given below are two statements :
Statement-I : In first phase of protein synthesis (translation) amino acids are activated in the presence of ATP.
Statement-II : Formation of peptide bond requires energy.
In the light of the above statements, choose the correct answer from the options given below :

(1) Both Statement I and Statement II are incorrect.

(2) Statement I is correct but Statement II is incorrect.

(3) Statement I is incorrect but Statement II is correct

(4) Both Statement I and Statement II are correct.

16) Which of the following option is incorrect about *E.coli* ?

(1) *E.coli* has 4.6×10^6 bp of DNA

(2) Process of DNA replication completes in 18 minutes

(3) Rate of polymerisation by DNA polymerase is approximately 1000 bp/minutes

(4) In *E.coli* DNA replicate in semiconservative manner

17) Which of the following is not included in transcription unit ?

(1) Promoter

(2) Terminator

(3) Structural gene

(4) Co-repressor

18) Promoter and terminater are located at transcription unit :-

(1) Down stream and upstream respectively from structural gene

(2) Upstream and down stream respectively from structural gene

(3) Both upstream from structural gene

(4) Both down stream from structural gene

19) What is translation unit?

(1) Sequence on DNA that is flanked by promotor and terminator

(2) Sequence on mRNA that is flanked by starting and stop codons

(3) Sequence on DNA that is flanked by starting and stop codons

(4) Sequence on mRNA that is flanked by promotor and terminator

20) In the Griffith experiment following representation A and B is - S Strain → injected into Mice → A
R Strain → injected into Mice → B

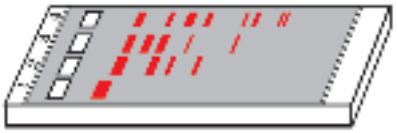
(1) A → Mice die
B → Mice live

(2) A → Mice die
B → Mice die

(3) A → Mice live
B → Mice live

(4) A → Mice live
B → Mice die

21) According to following diagram 'DNA' appears in the form of _____ and stained by _____



(1) Bright orange colour, Janus Green B

(2) Bright orange colour, Ethidium bromide

(3) Bright blue colour, Janus Green B

(4) Bright yellow colour, Ethidium bromide

22) What will be the sequence of mRNA produced by the following stretch of DNA?
3' A T G G A A C T A 5' Template strand
5' T A C C T T G A T 3' Coding strand

(1) 3' A T C C G A A C U 5'

(2) 5' U A C C U U G A U 3'

(3) 5' A U G G A A C U A 3'

(4) 3' A U A A G G C U A 5'

23) Which of the following enzyme helpful in polymerising RNA in a template indepent manner ?

(1) Severo ochoa enzyme

(2) DNA polymerase

(3) RNase

(4) Ligase

24) Which of the following statement is incorrect with respect to t - RNA ?

(1) t RNA has an anticodon loop

(2) t RNA has an amino acid acceptor end

(3) t RNA is an adapter molecule

(4) t RNA act as a catalyst

25) Which statement is correct for transcription ?

(1) Both strands of DNA are copied during transcription

(2) Promoter region is said to be located toward 3' end (downstream) of structural gene.

(3) Whole DNA of an organism gets transcribed.

(4) Sigma factor helps RNA polymerase to bind with promoter region.

26) **Statement-I** : RNA has evolved from DNA with chemical modifications that make RNA more stable.
Statement-II : DNA being double stranded and having complementary strand further resists changes by evolving process of repair.

(1) Both statement-I and statement-II are true.

(2) Both statement-I and statement-II are false.

(3) Statement-I is true but statement-II is false.

(4) Statement-I is false but statement-II is true .

27) Match the column:

Column-I		**Column-II**	
(A)	Triplet	(i)	One codon codes for only one amino acid
(B)	Non-ambiguity	(ii)	One amino acid is coded by more than one codon
(C)	Universality	(iii)	A codon is made up of three N_2 bases.
(D)	Degeneracy or redundancy	(iv)	From bacteria to human beings UUU codes for phenyl alanine

(1) A = (iii), B = (ii), C = (iv), D = (i)

(2) A = (iii), B = (i), C = (iv), D = (ii)

(3) A = (iv), B = (i), C = (ii), D = (iii)

(4) A = (iii), B = (iv), C = (i), D = (ii)

28) Which one of the following pairs is correctly matched ?

(1) Ribosomal RNA – Carrier of amino acid

(2) Transcription – Process by which protein is synthesized

(3) Replication – Process by which m-RNA carries the information from nucleus to the ribosome

(4) Anticodon – Site of t-RNA molecule that contains complementary bases to the triplet code on the m-RNA

29) Select the **incorrect** statement(s) :-
(A) Six-codons do not code for any amino acid
(B) Codons are read in m-RNA in a contiguous fashion
(C) Three codons function as stop codons
(D) The initiator codon AUG codes for methionine

(1) A and B

(2) B, C and D

(3) A and C

(4) Only A

30) Find out correct match from column-I and II

Column-I		**Column-II**	
a	Monosomy	i	2n –1
b	Double monosomy	ii	2n + 1
c	Nullisomy	iii	2n –1 – 1
d	Trisomy	iv	2n – 2

(1) a-iii, b-ii, c-i, d-iv

(2) a-i, b-iii, c-iv, d-ii

(3) a-iv, b-ii, c-iii, d-i

(4) a-iii, b-iv, c-ii, d-i

31)

In lac operon the repressor protein bind at which region of operon.

(1) Promotor

(2) Operator

(3) Z-Gene

(4) Y-Gene

32) **Statement I :** Origin of replication is a definite region in *E.Coli* DNA where the replication originates.
Statement II : During transcription RNA polymerase bind at promoter region of DNA.
Choose the correct answer :

(1) Both statement I and statement II are true

(2) Both statement I and statement II are false

(3) Statement I is correct but statement II is false

(4) Statement I is false but statement II is correct

33) *E.Coli* DNA has only____I____ base pairs and the process of replication complete with in ____II____ minutes.
Choose the correct data for I and II blank.

	I	II
1	6.6×10^9	38
2	48 502	18
3	4.6×10^6	18

4	3.3×10^9	38

(1) 1

(2) 2

(3) 3

(4) 4

34) Association with which of the following factor alter the specificity of the RNA polymerase to terminate the transcription ?

(1) Release Factor

(2) Sigma Factor

(3) Rho Factor

(4) Initiation Factor

35) Which of the following statement is incorrect ?

(1) In bacteria the transcription and translation can be coupled.

(2) Nirenberg's cell free system for protein synthesis helped the code to be deciphered.

(3) AUG has dual function

(4) The UTR is present at both 5' end and 3' end of DNA.

36)

hn RNA subjected to a process called splicing where the ___I____ are removed and ____II____ are joined in a defined order.
Choose the correct word for I and II blank.

	I	II
1	Start codon	Stop codon
2	Exon	Intron
3	Intron	Exon
4	Stop codon	Start Codon

(1) 1

(2) 2

(3) 3

(4) 4

37) **Statement I :** The terminator is located towards 3' end of the coding strand.
Statement II : GGG and GGC both codons code for the amino acid Glycine.
Choose the correct option

(1) Both statement I and statement II are true

(2) Both statement I and statement II are false

(3) Statement I is true and statement II is false

(4) Statement I is false and statement II is true

38) Which of the following is a real inducer of lac operon ?

(1) Allolactose
(2) Glucose
(3) Fructose
(4) Ribose

39) Which of the following is not a correct statement?

(1) Genetic code is nearly universal
(2) Promotor is present on mRNA
(3) Codon read in contiguous fashion
(4) UAA act as a stop codon

40) Which of the following is not a correct statement

(1) In secondary structure tRNA looks like a colver leaf
(2) RNA used to act as a genetic material
(3) Polycistronic condition mostly found in eukaryotes
(4) In capping 7 methyl guanosine triphosphate is added.

41)

In lac operon the repressor protein binds to the ___I___ region of the operon and prevents ___II___ from transcribing the operon.
Choose the correct word for I and II blank :

	I	II
1	Operator	RNA Polymerase
2	Promotor	RNA Polymerase
3	Inducer	β - galactoside
4	Satellite	RNA Polymerase

(1) 1
(2) 2
(3) 3
(4) 4

42) Which of the following is the substrate for the exzyme Beta - galactosidase and it also regulates switching on and off of the lac operon.

(1) Lactose
(2) Glucose
(3) Galactose

(4) Distrophin

43) Match the column I with column II

	Column I		**Column II**
a	Sickle cell anemia	i	Trisomy
b	Down syndrome	ii	Point mutation
c	Cancer cells	iii	Mutagen
d	UV radiations	iv	Chromosomal aberrations

(1) a - iv, b - i, c - iii, d - ii

(2) a - ii, b - iv, c - i, d - iii

(3) a - ii, b - i, c - iv, d - iii

(4) a - ii, b - iii, c- i, d - iv

44) During process of translation the ribozyme act as a ___I___ for the formation of ___II___ bond. Choose the correct word for I and II blank.

	I	II
1	Catalyst	Peptide
2	Release factor	Peptide
3	Rho factor	N-Glycosidic
4	Adapter	Phosphodiester

(1) 1

(2) 2

(3) 3

(4) 4

45) **Statement I :** A very low level of expression of lac operon has to be present in the cell all the time.
Statement II : In lac operon repressor protein show negative regulation.

(1) Statement I is true but statement II is false

(2) Both Statement I and Statement II are true.

(3) Both Statement I and Statement II are false

(4) Statement I is false but statement II is true.

46) Umblical cord is a connection in between

(1) Foetus and placenta

(2) Foetus and mother

(3) Placenta and mother

(4) Foetus and mother heart

47) Choose the incorrect statement :-

(1) The menstrual phase is followed by the follicular phase.
(2) The ovulation is followed by the luteal phase
(3) Disintegration of the endometrium leads to menstruation
(4) Follicular phase is followed by the menstrual phase.

48) In a normal, Healthy 25 year old woman with a menstrual cycle of 28 days :-

(1) The secretory nature of the uterus is develop by progesterone produced by corpus luteum
(2) Menstruation is caused by progesterone secrete from the corpus luteum.
(3) Injections of estrogens and progesterone will cause an inlargement of the ovaries and an increase in the production of mature ovarian follicles.
(4) The concentration of estradiol in the blood plasma begins to fall prior to ovulation and continous to decrease until menstruation.

49) Pregnancy is maintened by :-

(1) HCG
(2) LH
(3) Estrogen
(4) Progesterone

50) The main reason behind LH surge of 6-10 folds, while FSH surge of only 2-3 folds during preovulatory phase of menstrual cycle is

(1) high level of estrogen promotes release of FSH and LH
(2) Inhibin inhibits release of FSH and to a lesser extent of LH
(3) LH surge is necessary for ovulation
(4) both (1) and (2) simultaneously

51) The signals for parturition originate from :-
(A) Fully developed foetus
(B) Placenta
(C) Uterus
(D) Ovary

(1) A and B
(2) A and D
(3) C and D
(4) B and D

52) Hysterectomy is surgical removal of :

(1) Prostate gland
(2) Vas-deferense
(3) Mammary glands

(4) Uterus

53) Ectopic pregnancies are referred to as :

(1) Pregnancies terminated due to hormonal imbalance
(2) Pregnancies terminated due to hormonal imbalance
(3) Implantation of embryo at site other than uterus.
(4) Implantation of defective embryo in the uterus

54) Around 14th day of ideal menstrual cycle LH increases 6-10 times but FSH increases only 2-3 times due to :-

(1) Positive feedback of inhibin
(2) Negative feedback of inhibin
(3) Positive feedback of progesterone
(4) Negative feedback of estrogen

55) Extraembryonic membranes which participate in the formation of placenta in human are:

(1) Chorion and Allantois
(2) Chorion and Amnion
(3) Amnion and Allantois
(4) Amnion and Yolk sac

56)

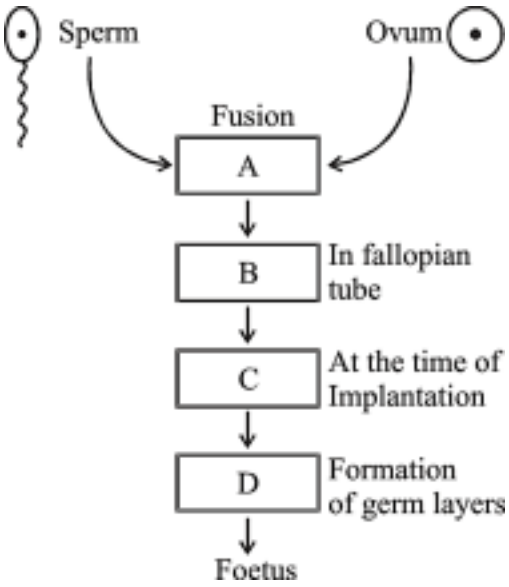
In a female undergoing hysterectomy, which of the following event will not occur?

(1) Completion of meiosis-II and formation of ovum
(2) Fertilization
(3) Menstruation
(4) All of these

57) Menstruation is initiated by

(1) A sudden release of FSH from the anterior pituitary
(2) A lack of estrogen and progesterone due to degeneration of the corpus luteum
(3) An increased release of estrogen and progesterone from the corpus luteum
(4) A sudden drop in FSH

58) Study the cycle shown below and select the option which gives correct words for all four blanks



A, B, C, D :-

	A	B	C	D
(1)	Foetus	Zygote	Morula	Gastrula
(2)	Zygote	Morula	Blastocyst	Gastrula
(3)	Embryo	Blastocyst	Gastrula	Cleavage
(4)	Zygote	Morula	Gastrula	Cleavage

(1) 1

(2) 2

(3) 3

(4) 4

59) Identify the contraceptive device shown below as well as the related right place of its



implantation into a woman and select the correct option together :

	Contraceptive device	Site of implant
(1)	LNG – 20	Fallopian tube
(2)	Lippes loop	Uterine wall
(3)	Copper–T	Uterine cavity

(4)	Multiload 375	Oviduct

(1) 1

(2) 2

(3) 3

(4) 4

60) **Statement-I :** Tertiary follicle is characterised by a fluid filled cavity called antrum.
Statement-II : Primary oocyte completes its first meiotic division in secondary follicle.

(1) **Statements I & II** both are correct

(2) **Statements I & II** both are incorrect

(3) Only **statement-I** is correct

(4) Only **statement-II** is correct

61) **Statement-I :** The follicles surrounded by more layers of granulosa cells and a new Theca are called secondary follicle.
Statement-II : In tertiary follicle the theca folliculin is clearly organised into an inner theca interna and an outer theca externa.

(1) Only **statement-I** is correct

(2) Only **statement-II** is correct

(3) **Statements I & II** both are correct

(4) **Statements I & II** both are incorrect

62) **Statement-I :** After fusion of the male and female gametes the zygote would carry either XX or XY. Its depending whether ovum carrying X or Y.
Statement-II : The blastocyst embedded in the endometrium of uterus this is called implantation.

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I and Statement II both are incorrect.

(3) Statement I is incorrect but Statement II is correct.

(4) Both Statement I and Statement II are correct.

63) Which of the following statement is incorrect ?

(1) The trophoblast contains certain cells called stem cells.

(2) The trophoblast layer gets attached to the endometrium.

(3) A mesoderm appears between the ectoderm and the endoderm.

(4) Three layers give rise to all tissues in adults.

64) How many of following hormones are produced in women only during pregnancy ?
HCG, HPL, Relaxine, estrogens, progestogens, thyroxine.

(1) Three

(2) Four

(3) Five

(4) Two

65) Given below are two statements :-
Statements-I :- Oxytocin acts on the uterine muscles and causes stronger uterine contractions.
Statements-II :- Vigorous contraction of the uterus at the end of pregnancy causes delivery of the foetus.

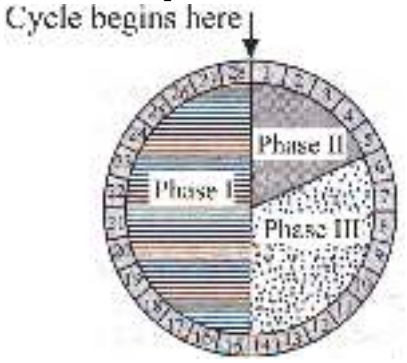
(1) Both statement I and II are correct.

(2) Both statement I and II are incorrect.

(3) Statement I is correct but Statement II is incorrect

(4) Statement I is incorrect but statement II is correct.

66) Refer the given schematic representation of menstrual cycle showing three phases I, II and III. Select the option that correctly matches these phases with the following events :-



(i) FSH secreted by the anterior pituitary stimulates the ovarian follicle to secrete oestrogen which stimulates the proliferation of endometrium.
(ii) LH secreted by the anterior pituitary stimulates the development of corpus luteum which secretes progesterone
(iii) Reduced production of LH from anterior pituitary causes degeneration of corpus luteum, thereby menustral flow starts.

(1)	Phase - I (i)	Phase - II (ii)	Phase - III (iii)
(2)	Phase - I (iii)	Phase - II (i)	Phase - III (ii)
(3)	Phase - I (ii)	Phase - II (iii)	Phase - III (i)
(4)	Phase - I (iii)	Phase - II (ii)	Phase - III (i)

67) Which of the following hormone responsible for milk formation in Human female ?

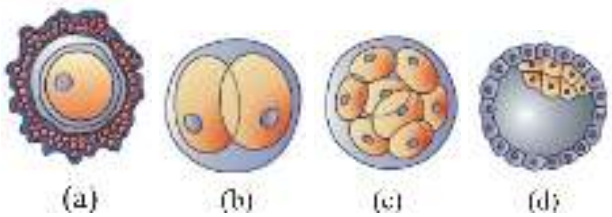
(1) Prolactin

(2) Oxytocin

(3) Estrogen

(4) Progesterone

68) The figure below show four stage (a, b, c, d) of human development. Select the option giving correct identification together with site of occurrence ?



		Developmental stage	Site of occurrence
(1)	(a)	Fertilised egg	Isthms part of fallopian tube
(2)	(b)	Zygote	Distal part of fallopian tube
(3)	(c)	Morula	Starting point of fallopian tube
(4)	(d)	Blastocyst	Uterine cavity

(1) 1
(2) 2
(3) 3
(4) 4

69) During menstrual cycle which hormone attains two peaks ?

(1) Progesterone
(2) Estrogen
(3) FSH
(4) LH

70) The first sign of growing foetus may be noticed by:-

(1) Movement of fetus
(2) Appearance of hair on head
(3) Listening to the heart sound carefully through the stethoscope
(4) Formation of limbs

71) Rupture of mature graffian follicle to release ovum is caused by :-

(1) LH
(2) FSH
(3) GnRH
(4) Estrogen

72) Match the Column–A with Column–B :-

Column-A		Column-B	
(A)	Follicular phase	(i)	Release of secondary oocyte

(B)	Luteal phase	(ii)	Fully mature graafian follicle
(C)	Menstrual phase	(iii)	Formation of corpus luteum
(D)	Ovulatory phase	(iv)	Breakdown of endometrium

(1) A–(i), B–(ii), C–(iv), D–(iii)

(2) A–(ii), B–(iii), C–(i), D–(iv)

(3) A–(ii), B–(iii), C–(iv), D–(i)

(4) A–(iv), B–(ii), C–(i), D–(iii)

73) Which of the following contraceptive doesn't make uterus unsuitable for implantation ?

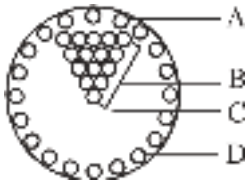
(1) LNG-20

(2) Progestasert

(3) Today

(4) Saheli

74) Identify the structure and its labelling in the following diagram correctly -



		A	**B**	**C**	**D**
(1)	Blastocyst	Trophoblast	Embryonal, Knob	Blastocoel	Cells of Rauber
(2)	Gastrula	Cells of Rauber	Embryonal, Knob	Gastrocoel	Trophoblast
(3)	Blastocyst	Cells of Rauber	Embryonal, Knob	Blastocoel	Trophoblast
(4)	Blastocyst	Cells of Rauber	Inner cell mass	Blastocoel	Zona pellucida

(1) 1

(2) 2

(3) 3

(4) 4

75) Which of the following methods of contraception has least side effects ?

(1) IUD

(2) Pills

(3) Coitus interruptus

(4) Cervical cap

76) All are the sexually transmitted diseases except :–

(1) Syphillis
(2) Gonorrhoea
(3) Hepatitis–B
(4) Polio

77) Which of the following approaches does not give the defined action of contraceptive ?

(1)	Barrier methods	prevent fertilization
(2)	Intra uterine devices	Increase phagocytosis of sperms, suppress sperm motility and fertilizing capacity of sperms
(3)	Hormonal contraceptives	Prevent/retard entry of sperms, prevent ovulation and fertilization
(4)	Vasectomy	Prevents spermatogenesis

(1) 1
(2) 2
(3) 3
(4) 4

78) Match the column–I with column–II :-

Column-I		**Column-II**	
(A)	Copper releasing IUDs	(i)	Inhibit ovulation
(B)	Oral pills	(ii)	Suppress sperms motility
(C)	Sterilisation methods	(iii)	Make the uterus unsuitable for implantation
(D)	Hormonal releasing IUDs	(iv)	Block gametes transport

(1) A-iii, B-ii, C-i, D-iv
(2) A-iv, B-i, C-ii, D-iii
(3) A-ii, B-i, C-iv, D-iii
(4) A-ii, B-iii, C-i, D-iv

79) If sperm is immotile which of the following technique is suitable for fertilization :-

(1) Z.I.F.T
(2) I.U.T.
(3) I.C.S.I.
(4) A.I.

80) Lactating mother generally does not concieve because of -

(1) Hypersecretion of gonadotropins

(2) Suppression of Gonadotropins secretion

(3) Suppression of fertilization

(4) Suppression of gamete transport

81) MTP is illegally used :-
(A) To prevent ovulation.
(B) For female foeticide.
(C) When continuation of the pregnancy could be harmful to the mother.
(D) For determination of sex of unborn child.

(1) B and D

(2) B, C and D

(3) Only B

(4) A, B and D

82) Which of the following technique is banned in india to prevent female foeticides :-

(1) IVF

(2) Ultra sound

(3) Amniocentesis

(4) MRI

83) Consider the statements given below regarding contraception and answer as directed there after :-
(A) Saheli prevent implantation.
(B) Generally chances of conception are nil until mother breast feeds the infant up to two year.
(C) Intrauterine devices like copper T are natural contraceptive
(D) Chances of fertilisation are very high during first 7 days of menstruation cycle.
How many of the above statements are incorrect?

(1) 2

(2) 1

(3) 4

(4) 3

84) Government sponsored "family planning programme" started in:-

(1) 1947

(2) 1951

(3) 1977

(4) 1955

85) Which of the following is mis-matched ?

(1) Trichomoniasis - *Trichomonas vaginalis*

(2) Genital Herpes - Herpes simplex virus

(3) Gonorrhoea - *Trepanoma pallidum*

(4) Genital warts - Human papilloma virus

86) **Statement I :-** Stellar distances are measured in miles.
Statement II :- The universe is very old, almost 20 billion years old.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct II is incorrect.

(4) Statement I is incorrect II is correct.

87) In the solar system of the milky way galaxy, earth was supposed to have been formed about _____ years back. Choose the correct data among given options to fill the blank space.

(1) 4 Billion

(2) 4.5 Billion

(3) 0.5 Billion

(4) 2 Million

88) **Statement I :-** The big bang theory talks of a singular huge explosion unimaginable in physical terms.
Statement II :- Huge cluster of galaxies comprise the universe. Galaxies contain stars and clouds of dust and gas.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct II is incorrect.

(4) Statement I is incorrect II is correct.

89) Life appeared __A__ million years after the formation of earth, i.e. almost ___B___ billion years back complete the given statement by choosing choosing correct A and B from option.

(1) A → 500 ; B → Four

(2) A → 400 ; B → Four

(3) A → 200 ; B → Two

(4) A → 300 ; B → Five

90) **Assertion (A) :-** There was no atmosphere on early earth.
Reason (R) :- Oxygen combined with ammonia and methane to form water.

(1) Both 'A' and 'R' are correct and 'R' is the correct explanation of 'A'.

(2) Both 'A' and 'R' are correct and 'R' is not correct explanation of 'A'.

(3) A is correct but R is incorrect

(4) A is incorrect but R is correct

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	4	2	4	1	3	1	2	2	1	3	4	1	3	2	2	1	4	3	3	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	4	4	2	2	2	2	3	4	1	3	4	4	1	3	1	3	2	1	3
Q.	41	42	43	44	45															
A.	3	2	1	3	1															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	1	4	1	3	2	1	1	3	2	4	3	3	2	3	4	4	4	1	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	2	3	3	3	2	4	3	2	3	2	4	3	2	3	2	3	1	3	1	4
Q.	86	87	88	89	90															
A.	1	4	4	2	4															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	1	3	3	2	1	2	1	1	3	1	4	3	2	4	4	3	4	2	2	1
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	2	1	4	4	4	2	4	4	2	2	1	3	3	4	3	1	1	2	3
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	1	3	1	2	1	4	1	4	4	1	4	3	2	1	4	2	2	3	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	3	1	1	1	3	1	4	2	3	1	3	3	3	3	4	4	3	3	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	3	4	2	3	2	2	1	1	2										

SOLUTIONS

PHYSICS

1)

Question Asking About:
We need to determine if the focal length of a spherical mirror changes with the color (wavelength) of light.

Key Concept:
The focal length of a spherical mirror is defined by its geometry, specifically its radius of curvature.

Formula f = R/2

Solution/Explanation/Calculation:
where R is the radius of curvature. This formula shows that the focal length depends only on the radius of curvature, which is a physical property of the mirror and does not depend on the wavelength of light.
The focal length of a spherical mirror is the same for light of all colors.

Conclusion:
Hence, option (4) is correct.

2) R = 25 cm
f = 12.5 cm

$$m = \frac{f}{f-u} = \frac{12.5}{12.5-(-40)}$$

m = 0.23

$$m = \frac{\text{height of image}}{\text{height of object}}$$

height of image = 0.23 × 7.5 = 1.78

3)

$$m = \frac{A_i}{A_0} = \left(\frac{f}{f-4}\right)^2$$

$$\frac{A_i}{l^2} = \left(\frac{15}{15-20}\right)^2$$

$A_i = 9\ cm^2$

4)

for point A
u = – 30 ; f = – 10

$v_1 = \frac{-30(-10)}{-30-(-10)}$ = – 15 cm

for point B

u = – 35 ; f = – 10

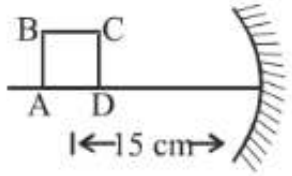
$v_2 = \frac{-35(-10)}{-35-(-10)}$ = –14 cm

$L_I = v_2 - v_1$ = – 14 – (– 15) = 1 cm

5)

Question Explanation:

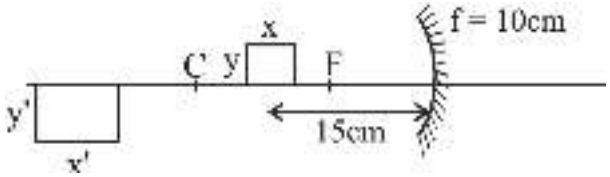
The question asks for the perimeter of the image of a square (side 1 mm) placed 15 cm in front of a concave mirror with a 10 cm focal length.



Concept: Longitudinal magnification for very small length

Solution:

By transverse magnification



$$\frac{y'}{y} = \frac{f}{f-u}$$

Given :

f = –10cm

u = –15 cm

y = 1 mm

$y' = \left(\frac{-10}{-10+15}\right)$ (1 mm) = –2mm

(y') = 2mm

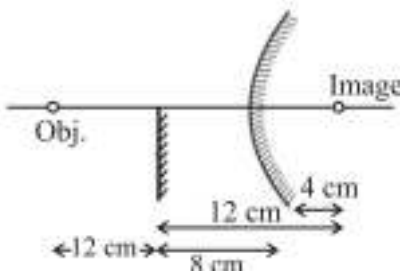
x is very very small hence use longitudinal magnification formula for very small length .

$\frac{x'}{x} = m^2 \Rightarrow x' = m^2 x = \left(\frac{f}{f-u}\right)^2$ x = 4mm

Perimeter of image

= 2x' + 2y' = (2 × 4) + (2 × 2) = 12 mm

Final Answer : Option (3)



6)

for spherical mirror u = –20 cm

v = 4 cm

$$\frac{1}{u}+\frac{1}{v}=\frac{1}{f}$$

$$-\frac{1}{20}+\frac{1}{4}=\frac{1}{f} \Rightarrow \frac{1}{f}=\frac{1}{5}$$

f = +5 cm

7)

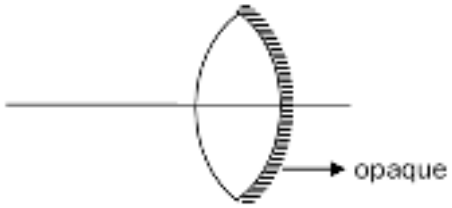
Question Explanation:

Determine the effect on an image formed by a concave mirror when half of the mirror is covered.

Concept:

Every part of a mirror forms a complete image, but the amount of light reflected determines the intensity.

Solution:



Each point on the mirror receives light from every point of the object and reflects it to the corresponding image point.

Covering the lower half of the mirror blocks half of the light rays contributing to the image.

The geometry of the rays forming the image remains unchanged, so the full image is still formed at the same location.

The total amount of light energy reaching the screen is reduced by half.

This results in a decrease in the brightness (intensity) of the image.

Final answer: Option (2)

8) Velocity of image with respect to object is twice of relative velocity between mirror and object.

9)

Question Explanation: .

The question asks how fast the image moves when an object is moving towards a concave mirror along its principal axis. The object is 20 cm from the mirror, the focal length is 15 cm, and the object moves at 5 cm/s.

Concept: velocity of image in spherical mirror
Solution:

$$m = \frac{f}{f-u} = \frac{-15}{-15+20} = \frac{-15}{5} = -3$$

$V_I = -m^2 V_O$

$V_I = -(-3)^2 (5) = -45$ cm/s

Final answer : option (1)

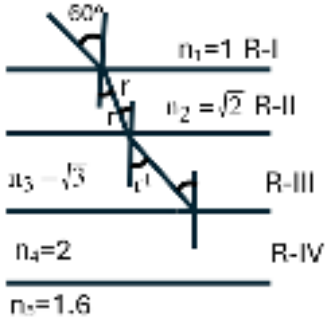
10) $f = \sqrt{x_1 x_2} = \sqrt{16 \times 9} = 12\text{cm}$

11) **A. Question Explanation :** We need to find the relationship between the refractive indices.
B. Concept : When a ray light passes through multiple parallel interfaces, the final emergent ray is parallel to the incident ray if the first and last media have the same refractive index. This is because the net deviation is zero.
C. Formula : Snell's law : $n_1 \sin\theta_1 = n_2 \sin\theta_2$
D. Calculation : Since the surfaces are parallel and the emergent ray CD is parallel to the incident ray AB, it means that the net deviation of the light ray is zero. This can only happen if the first and last media have the same refractive index. Therefore, $\mu_4 = \mu_1$
E. Final answer : 4

12) **Explanation:**
To find out angle of incident with normal in medium of refractive index of $\sqrt{3}$ as per given conduction.
Concept based:
snell's law and refraction by plance surface.

Formula:
Snell's law

$$\frac{\sin_1}{\sin_r} = \frac{\mu_2}{\mu_1}$$



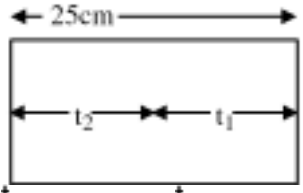
By diagram and snell's law

$$R_I \to R_{II} \frac{\sin 60^o}{\sin r} \frac{\sqrt{2}}{1} \Rightarrow \sin r^r = \frac{\sqrt{3}}{2\sqrt{2}}$$
$$R_{II} \to R_{III} \frac{\sin r}{\sin r^1} = \frac{\sqrt{3}}{\sqrt{2}} \Rightarrow \sin r^1 = \frac{1}{2}$$
$\Rightarrow r^1 = 30°$

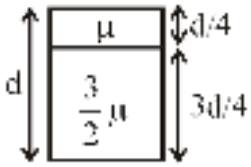
13)

Explain the Question : Refractive index of glass w.r.t oil is 3/5.
Absolute indices: μg = 3/2 and μw = 4/3.
We must find refractive index of water w.r.t oil.
Concept : This question is based on refractive index between two media.
Formula :
$\mu_{(a/b)} = \mu_a/\mu_b$.
Solution :
Given :
$\mu_{(g/o)} = 3/5$
$\mu_g = 3/2$
Using $\mu_{(g/o)} = \mu_g/\mu_o$
$3/5 = (3/2)/\mu_o$
$\mu_o = (3/2) \div (3/5)$
$= (3/2) \times (5/3)$
$= 5/2$
Now required :
$\mu_{(w/o)} = \mu_w/\mu_o$
$= (4/3) \div (5/2)$
$= (4/3) \times (2/5)$
$= 8/15$
Final answer : 3

14) **Explain the Question :** To find refractive index of the slab.
Concept : This question is based on normal shift.
Solution :



$\frac{t_1}{\mu} = 6$, $\frac{t_2}{\mu} = 4$
$t_1 = 6\mu$, $t_2 = 4\mu$
$t_{net} = t_1 + t_2 = 25$
$= 6\mu + 4\mu = 25$
$10\mu = 25$
$\mu = 2.5$
Final Answer : 2



15)

$$d_{a_{PT}} = d_{a_{P_1}} + d_{a_{P_2}}$$

$$\frac{d}{2} = \frac{d}{4\mu} + \frac{3}{4} \times \frac{2d}{3\mu} = \frac{d}{4\mu}[1+2]$$

$$\Rightarrow \frac{1}{2} = \frac{3}{4\mu} \Rightarrow \mu = \frac{3}{2}$$

16) App. depth = $\frac{\text{real depth}}{\mu} = \frac{12.5}{1.5} = 8.3$

so shift is 9.4 – 8.3 = 1.1 cm

17) **Explain the Question :** To find the critical angle for surface seperating media A and B.

Concept : This question is based on critical angle and total internal reflection

Solution :

$$\sin\theta_C = \frac{V_{denser}}{V_{parer}}$$

$$\sin\theta_C = \frac{1.8 \times 10^8}{2.4 \times 10^8}$$

$$\sin\theta_C = \frac{18}{24} = \frac{3}{4}$$

$q_C = \sin^{-1}(3/4)$

Final Answer : 4

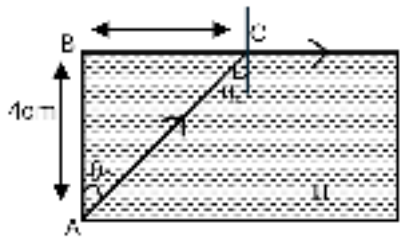
18) **Question**

To find the speed of light in liquid for given T.I.R condition.

Concept

TOTAL INTERNAL REFLECTION

VISUAL AID



Formula

For T.I.R $\sin\theta_C = \frac{1}{\mu}$ velocity of light $cm = \frac{c}{\mu}$

Solution / Calculation

According to figure $AC = 5cm$ $AC = \sqrt{AB^2 + BC^2}$

$\Rightarrow \sin\theta_c = \frac{BC}{AC} = \frac{3}{5}$

$\Rightarrow \sin\theta_c = \frac{1}{\mu}$

$\Rightarrow \frac{1}{\mu} = \frac{3}{5}$

speed of light in medium

$cm = 3 \times 10^8 \times \frac{3}{5}$

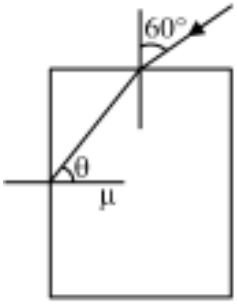
$= 1.8 \times 10^8 m/s$

Conclusion

Henec, option (3) is correct

19) **Explain the Question :** To find the minimum value of refractive index if T.I.R. occurs on surface BC **Concept :** This question is based on critical angle and TIR.

Solution : $\sin\theta \leqslant \sqrt{\mu^2 - 1}$



$\theta = 60°$

$\sin^2\theta \leq \mu^2 - 1$

$\sin^2 60° \leq \mu^2 - 1$

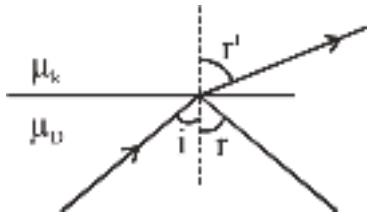
$\frac{3}{4} \leqslant \mu^2 - 1$

$\frac{3}{4} + 1 \leqslant \mu^2$

$\mu \geqslant \sqrt{\frac{7}{4}} \Rightarrow \mu \geqslant 1.32$

Final Answer : option (3)

20)



$r' + r + 90° = 180°$

$r' = 90° - r$

$\mu \sin i = 1 \times \sin r'$

$\mu \sin i = \sin(90° - r)$

$\mu = \frac{\cos r}{\sin i} = \cot i \, [i = r]$

$\frac{1}{\sin\theta_C} = \cot r \Rightarrow \sin\theta_C = \tan r$

$\Rightarrow \theta_C = \sin^{-1}(\tan r)$

21) **Question Explanation:** Rod of length f/3 lies on axis of concave mirror; one end of magnified image touches rod. Find image length.

Concept:
For touching condition in real magnified image, one end of rod is at centre of curvature.

Formula: C = 2f
$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ **Calculation:**
Rod length: $AB = \frac{f}{3}$ Place end A at centre of curvature: $u_A = 2f$ So: $v_A = 2f$ Other end: $v_B = 2f - \frac{f}{3} = \frac{5f}{3}$ Mirror formula: $\frac{1}{f} = \frac{1}{v_B} + \frac{1}{5f/3}$ $v_B = \frac{5f}{2}$ Image length: $L = v_B - v_A$
$L = \frac{5f}{2} - 2f = \frac{f}{2}$ **Final Answer:** Option (2)

22) **Concept :** Magnification

Formula : $M = \frac{f}{f - \mu}$

Calculation : $M_1 = \frac{f}{f + 25}$, $m_2 = \frac{f}{f + 40}$

$\frac{M_1}{M_2} = \frac{f + 40}{f + 25}$
$\frac{4}{1} = \frac{f + 40}{f + 25}$
f + 40 = 4f + 100
f = 20 cm
Final Answer : Option (4) 20 cm

23) $I = \int Fdt = \int_0^2 (24t^2 - 8t)dt$
$= \left[8t^3 - 4t^2\right]_0^2 = 64 - 16 = 48$ kg m/s

24) $\left|\vec{F}_{av}\right| = \frac{m\left[\sqrt{2gh_2} + \sqrt{2gh_4}\right]}{\Delta t} = \frac{50 \times 10^{-3}}{0.02}[8 + 10]$
= 45 N

25) $\vec{r} = (3 + 4t^2)\hat{i} + 2t\hat{j} + (3 - 6t)\hat{k}$
$\vec{v} = 8t\hat{i} + 3\hat{j} - 6\hat{k}$

$\vec{a} = 8\hat{i}$
as acceleration is along x-axis.
So net force will be along x-axis.

26) $F = \frac{dP}{dt} =$ Slope of P-t curve
So $\frac{F_A}{F_B} = \frac{\tan 30^\circ}{\tan 45^\circ} = \frac{1}{\sqrt{3}}$

27)

Area of f-t curve = ΔP
So $\Delta P = \frac{1}{2} \times 2 \times 10 + 2 \times 10 + \frac{1}{2} \times 30 \times 2$
$+ \frac{1}{2} \times 4 \times 20$
ΔP = 100
mv –0 = 100 ⇒ (v = 50 m/s)

28)

Question Explanation :

We need to find the ratio of tension T_1 and T_2 for equilibrium of the block as shown in figure.

Concept :

Equilibrium under tension force involving strings.

Solution :

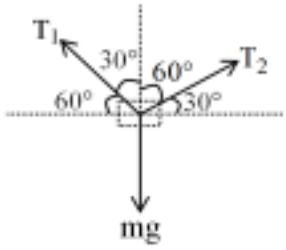
$$\frac{T_1}{\sin 120^o} = \frac{T_2}{\sin 150^o}$$

$$\frac{T_1}{T_2} = \frac{\sin 120^o}{\sin 150^o} = \frac{\sqrt{3}}{1}$$

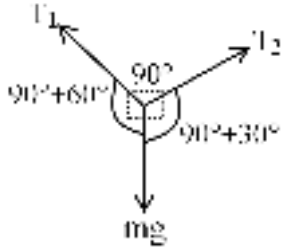
Final answer : Option (3)

29)

30 = (2 + 1) a, a = 10 m/s^2
F cos 60° = (4 + 3 + 2 + 1) × 10
$\frac{F}{2}$ = 100 N ⇒ F = 200 N



30)



Apply LAMI's Theorem

$$\frac{mg}{\sin 90^o} = \frac{T_1}{\sin(90+30)^o} = \frac{T_2}{\sin(90+60)^o}$$

$$\frac{mg}{1} = \frac{T_1}{\sin 60^o} = \frac{T_2}{\sin 30^o}$$

$T_1 = mg \sin 60°$

$\Rightarrow T_1 = mg\frac{\sqrt{3}}{2}$

$T_2 = mg \sin 30° = mg/2.$

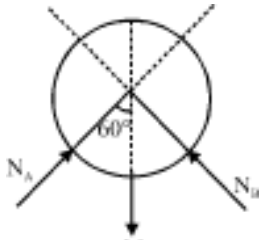
31)

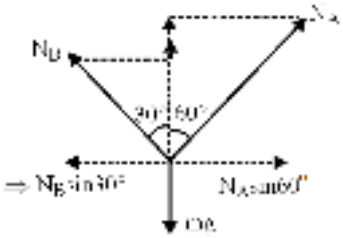
$N_1 = Mg$

$N_2 = Mg - 10\sin\theta$

$N_3 = Mg - 10$

$N_3 < N_2 < N_1$





32)

$N_A \sin 60° = N_B \sin 30°$

$N_A \cdot \frac{\sqrt{3}}{2} = N_B \frac{1}{2}$

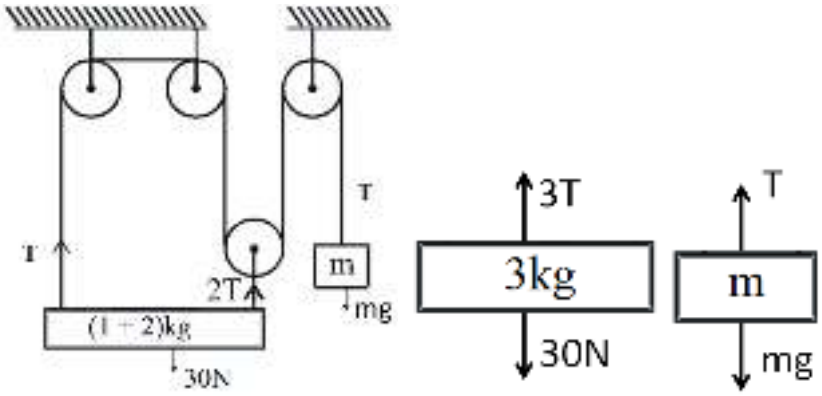
$\sqrt{3}N_A = N_B$

$\boxed{\sqrt{3}N_A = N_B}$

33) **Concept:** Equilibrium

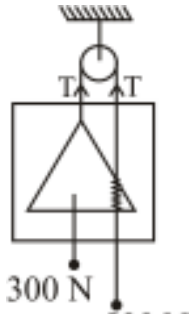
Formula : $F_{net} = 0$

Solution:



3T = 30 .... (1)
T = mg .... (2)
Put T in equation ---------- (1)
3mg = 30
m = 1kg

Final answer: Option (4)

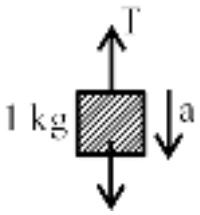


34)
At equilibrium
2T = 800
$\boxed{T = 400\,N}$

35)

Tension at rigid support.
T = 40(10 – 1) + 50(10 – 2) + 60(10 + 2)
= 40(9) + 50(8) + 60(12)
= 360 + 400 + 720
= 1480 N

36) $a = \frac{40-30}{7} = \frac{10}{7} m/s^2$
Now



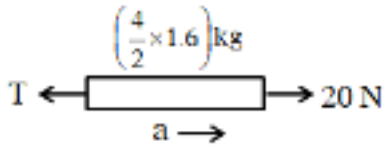
$10 - T = 1 \times \frac{10}{7}$

$T = 10 - \frac{10}{7} = \frac{60}{7} N$

37) $a = \frac{10}{2+4} = \frac{5}{3} m/s^2$

$kx = 2a = 2\left(\frac{5}{3}\right) = \frac{10}{3} N$

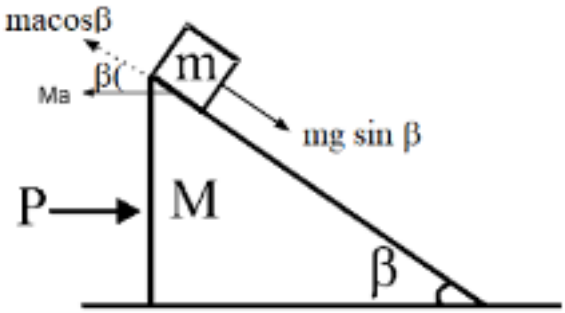
38) $a = \frac{20-8}{4} = 3\ m/s^2$



 $20 - T = 3.2 \times 3$

 $T = 20 - 9.6 = 10.4$ N

39)



For block m to remain stationary relative to M, the horizontal acceleration of M must match the component of gravity along the incline for m:

$a = g \sin \beta / \cos \beta$

$a = g \tan \beta$

Force on M

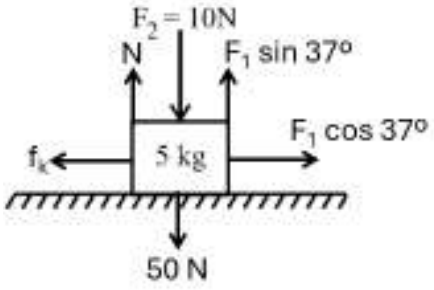
P = total mass × acceleration = (M + m) × g tan β

Final Answer: (1)

40) $a_{\text{Body lift}} = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)(g + a_0)$

$= \left(\frac{2-1}{2+1}\right)\left(g + \frac{g}{3}\right)$

$= \frac{1}{3}\left(\frac{4g}{3}\right) = \frac{4g}{9}$



41)

In vertical direction :

$N + F_1 \sin 37º = F_2 + 50$

$N + 60 \times \frac{3}{5} = 10 + 50$ ⇒ N = 24 Newton

In horizontal direction :

$F_{net} = F_1 \cos 37º - f_k$ (f_k = kinetic friction = $\mu_k N$)

$ma = 60 \times \frac{4}{5} - 0.5 \times 24$

$5a = 48 - 12$

$\Rightarrow a = \frac{48-12}{5} = \frac{36}{5} = 7.2\ m/s^2$

Final Answer: Option (3)

42)

for starting the motion

$F = f_{s_{max}}$

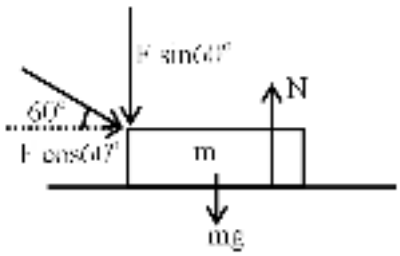
$4t = \mu_s mg$

$4t = 0.4 \times 10 \times 10$

$t = 10$ sec



Final Answer: option(2)



43)

in vertical direction

$N = mg + F \sin 60^o$

For motion

$F \cos 60^o \geqslant m[N]$

$F \times \frac{1}{2} \geqslant \frac{1}{2\sqrt{3}} [mg + F \sin 60^o]$

$$= \frac{F}{2} \geqslant \frac{1}{2\sqrt{3}} \left[\sqrt{3}g + \frac{\sqrt{3}F}{2} \right]$$

$$\frac{F}{2} \geqslant \frac{1}{2} \left[g + \frac{f}{2} \right]$$

$$F \geqslant g + \frac{F}{2} \Rightarrow \frac{F}{2} \geqslant 10 \Rightarrow 1F \geqslant 20N$$

44)

mg sin 37º ≤ μ(mg cos 37º + F)

$$20 \times \frac{3}{5} \leqslant 0.5 \left(20 \times \frac{4}{5} + F \right)$$

24 ≤ 16 + F

F ≥ 8 N

45)

m smooth inclined plane

$$S = \frac{1}{2}(g \sin 45^o)t^2$$

on rough inclined plane $a_{\text{Body wrt life}}$

$$S = \frac{1}{2}(g \sin 45^\circ - \mu g \cos 45^\circ) \times (2t)^2 \quad ...(2)$$

Now

$$\frac{1}{2}(g \sin 45^o)\, t^2 = \frac{1}{2}(g \sin 45^\circ - \mu g \cos 45^\circ) \times 4t^2$$

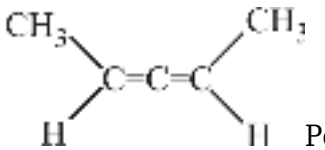
$$\frac{g}{\sqrt{2}} = \left(\frac{g}{\sqrt{2}} - \frac{\mu g}{\sqrt{2}} \right) \times 4$$

1 = (1 – μ)4

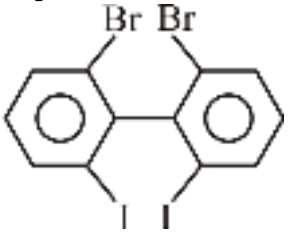
$$1 - \mu = \frac{1}{4} \Rightarrow \mu = 1 - \frac{1}{4} = \left(\frac{3}{4} \right)$$

CHEMISTRY

46) **Given Data:**



Statement I : Penta-2,3-diene is optically active due to perpendicular planes because of p-orbitals.



Statement II : Given o-substituted biphenyls are optically active due to the perpendicular plane because of the repulsion factor caused by bulky groups.

Concept :
Asymmetric molecules are known as chiral molecules. If plane of symmetry and center of chiral or not, check the plane of symmetry and center of symmetry..

Chiral molecules are optically active :

Answer Explanation :

Statement I : Penta-2,3-diene is chiral due to the perpendicular arrangement of the two double bonds. This perpendicularity prevents the molecule from having a plane of symmetry, which is a requirement for chirality. The p-orbitals are involved in the pi bonding of the double bonds and are essential for this perpendicular arrangement. Therefore, Statement I is correct.
Statement II : o-Substituted biphenyls can be chiral due to restricted rotation about the single bond connecting the two rings. Bulky groups at the ortho positions cause steric hindrance, forcing the rings into a perpendicular arrangement. This lack of free rotation and the resulting fixed perpendicular arrangement makes the molecule chiral. Therefore, Statement Il is correct.

Final Answer :
Option 1 (Both statement-I & statement-Il are correct)

47)

Question Explanation :

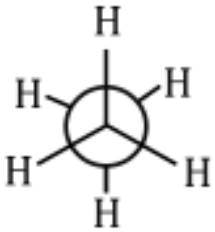
Identify the correct statement regarding stability of conformation of ethane.

Concept :

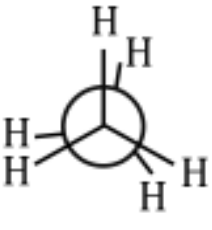
Conformation of ethane :- stability.

Solution :

The staggered conformation of ethane is more stable than eclipsed conformation. Since eclipsed conformation has a greater potential energy due to torsional strain, it is less stable than the staggered conformation.

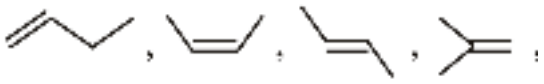


staggered

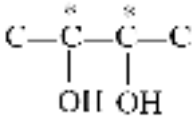


Eclipsed

Final answer : Option(1)



48) 



49)

carbon = 2 $p = \frac{n}{2} = 1$

$a = 2^{n-1} = 2^{2-1} = 2$

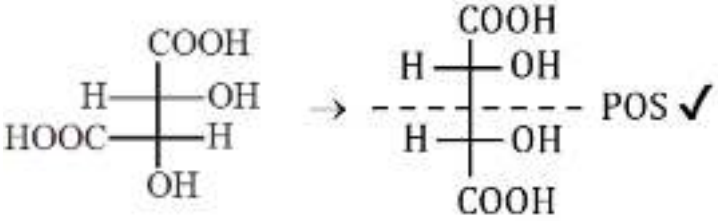
50) Homomers because both (A) and (B) have same R/S configuration.

51)

Question Explanation : Identify meso compounds.

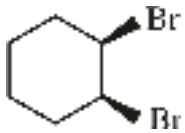
Concept : Meso compounds.

Solution :
Condition to become meso
(i) Atleast two chiral centre
(ii) Internal POS/COS

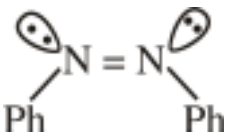


Chiral Center = 2, POS Present So, Meso Compound.

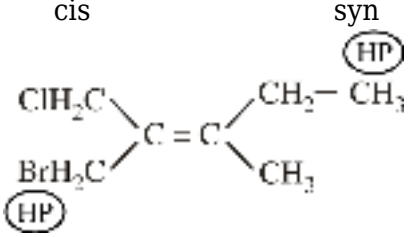
Final Answer : Option (2)

52) 

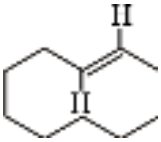
cis



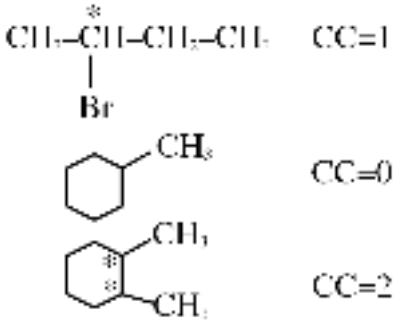
syn



(E)

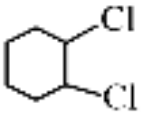


Trans

53) 

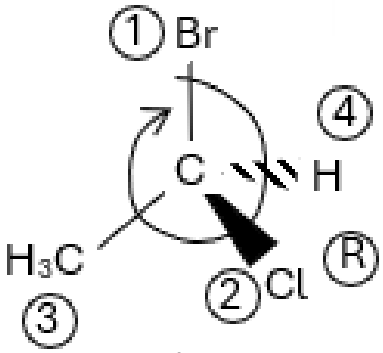
54)

compound without any C = C also show G.I.

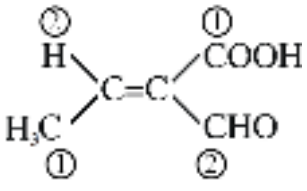


shows G.I.

55) **Concept:** R/S configuration.
• Follow CIP Rule.
• Give the priority to atoms/group on the basis of CIP Rule
Ans. Exp:-



Least priority (4th) at dash line, so we moves 1 → 2 → 3, we get dockwise direction
So. configuration is R.
Final Answer: (2)

56) 

E - isomer

57) **Explain Question:**
The question asks to identify which two structures among (I), (II), (III), and (IV) represent a pair of geometrical isomers.

Concept:

• Geometrical Isomerism (GI): Occurs due to restricted rotation (e.g., C = C).
• Condition: Each carbon atom of the double bond must be attached to two different groups.
• Isomer Relationship: Geometrical isomers must have the same molecular formula and same connectivity but different spatial arrangements.

Solution:
1. Check Condition:
• (i) has two Br atoms on one carbon. No GI.
• (III) has two H atoms on one carbon. No GI.
• (II) and (IV) both have different groups on each carbon (H/Cl and Br/CH3).
2. Compare (II) and (IV):
• Both share the formula C_3H_4ClBr.
• In (II), Cl and Br are cis (same side).
• In (IV), Cl and Br are trans (opposite sides).

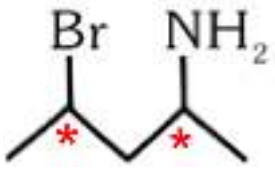
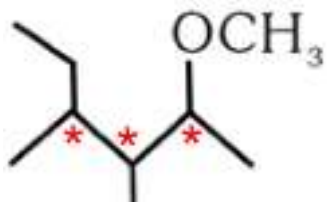
Final Answer: Option (3)

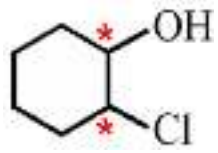
58)

Explain Question:
The question asks to identify the compound containing the maximum number of chiral centre's.

Concept:
Chiral centre's.

Solution:

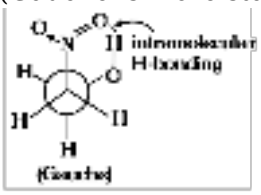
Given compounds	Chiral centre's
1 	2
2 	3
3 	4

4 	2

Final answer: option (3)

59)

$HO–CH_2–CH_2–NO_2$
(Gauche is more stable due to intramoleculer H-bonding)



60)

Question Explanation

The question asks which of the given pairs of compounds are NOT isomers. Isomers are molecules with the same molecular formula but different structural formulas (different arrangements of atoms in space).

Concept - To determine if two compounds are isomers, we need to:

A. **Determine the molecular formula of each compound.**
B. **Compare the molecular formulas.**
C. **If they have the same molecular formula, check if they have different structures.**

Answer Explanation - Let's analyze each pair:

A. **Propyne (C_3H_4) and Cyclopropene (C_3H_4):** Both have the same molecular formula but different structures (propyne is an alkyne, cyclopropene is a cyclic alkene). They are isomers.
B. **Propyne (C_3H_4) and Propadiene (C_3H_4):** Both have the same molecular formula but different structures (propyne has a triple bond, propadiene has two double bonds). They are isomers.
C. **Propene (C_3H_6) and Cyclopropene (C_3H_4):** Propene has the formula C_3H_6 (one double bond), and cyclopropene has the formula C_3H_4 (one double bond and a ring). They have *different* molecular formulas. Therefore, they are *not* isomers.
D. **1-Propanol (C_3H_8O) and Methoxyethane (C_3H_8O):** Both have the same molecular formula but different structures (1-propanol is an alcohol, methoxyethane is an ether). They are isomers.

Final Answer - 3 Propene and Cyclopropene

61) **Asking About -** Compounds show geometrical isomerism.

Concept -

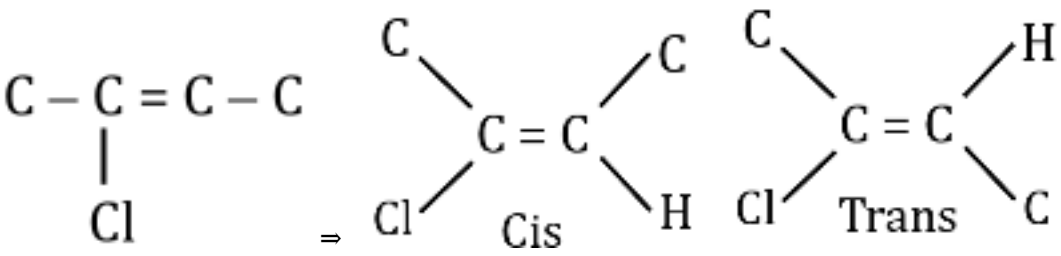
(1) Presence of restricted bond C=C; C=N; N=N; Cycloalkane; Cycloalkane

(2) Different group attached at each restricted atom.

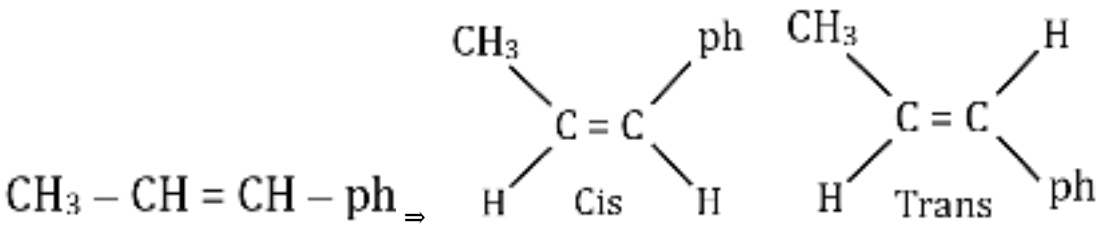
(3) Molecule should be planner.

Solution / Explanation -

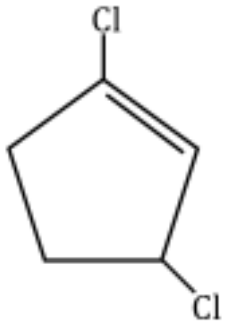
(a) 2–chloro but–2–ene



(b) 1–phenyl propene

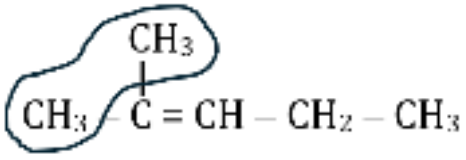


(c) 1, 3–dichloro cyclopentene

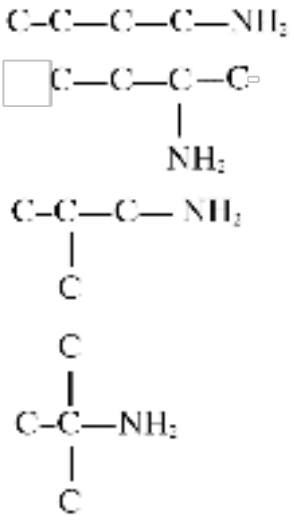


In cycloalkene 3 to 7 membered ring do not show geometrical isomerism across double bond.

(d) 2–methyl pent–2–ene

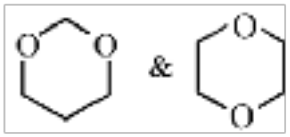
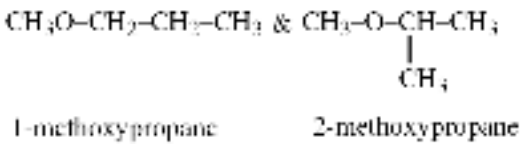
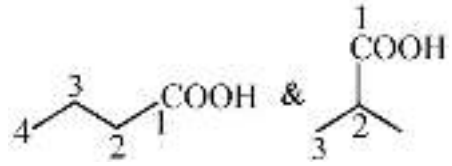


There group are same so, compound not show geometrical isomerism.

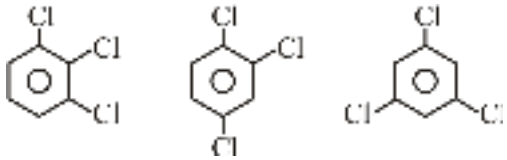
62) 

63)

Explain the Question : Determining which pair of compounds are not positional isomers.
Concept : This question is based on positional isomerism.
Solution :

	Given compounds	Remark
1	 Butan-1-ol Butan-2-ol	Positional isomers.
2	 1,3-dioxane 1,4-dioxane	Positional isomers or metamer.
3	 1-methoxypropane 2-methoxypropane	Positional isomers or metamer.
4		Both are chain isomers.

Final answer : 4

64) 

65)

C=C bond is also called Olefinic bond

66)

Question Explanation : The question asks to determine the total number of structural (constitutional) isomers for the molecular formula C_6H_{12} that specifically contain a four membered (cyclobutane) ring.

Concept : Structural Isomerism, Cycloalkanes, Degree of Unsaturation, Substituent Positioning.

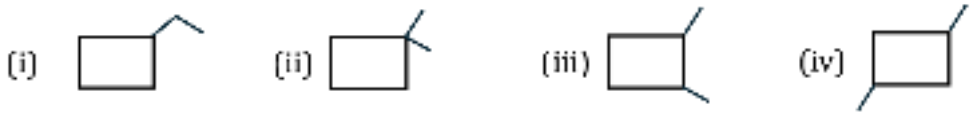
Solution : For C_6H_{12} with a 4 -membered ring, we have 2 remaining carbons to add as substituents:

(i) Ethylcyclobutane: A single ethyl group ($-CH_2CH_3$) attached to the ring.

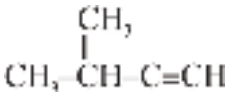
(ii) 1,1-Dimethylcyclobutane: Two methyl groups on the same carbon.

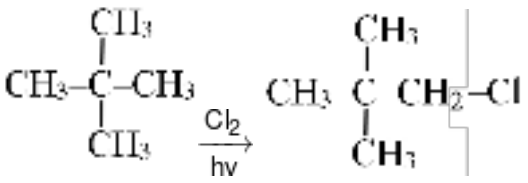
(iii) 1,2-Dimethylcyclobutane: Two methyl groups on adjacent carbons.

(iv) 1,3-Dimethylcyclobutane: Two methyl groups on opposite carbons.



Final Answer : Option (2)

67)  $CH_3–CH_2–CH_2–C\equiv CH$
chain isomers , in Alkyne min. 5C
$CH_3–CH_2–C\equiv CH$ $CH_3–C\equiv C–CH_3$
position isomers, in Alkyne min. 4C.

68) 

69) By observation we get this plot during measurable temperatures

Ans. 3rd Option.

70) **Explanation the Question:** The question provides the slope of a first-order reaction's Arrhenius plot (In k vs 1/T) and asks for the activation energy (E_a) in kcal/mol.
Concept: This question is based on Arrhenius Equation
Solution:

$$\text{In}(k) = \frac{E_a}{R}\left(\frac{1}{T}\right) + \ln(A)$$

$$\text{Slope (m)} = -\frac{E_a}{R}$$

We are given: Slope (m) = –4000 K

$$-4000\ \text{K} = -\frac{E_a}{R}$$

$E_a = 4000\ \text{K} \times R$

$E_a = 4000\ \text{K} \times 2\ \text{cal}\ \text{K}^{-1}\ \text{mol}^{-1}$

$$E_a = \frac{8000\,\text{cal/mol}}{1000\,\text{cal/kcal}} = 8\ \text{kcal/mol}$$

Final Answer: Option 2

71) Catalyst does not change value of ΔH.

72) Pre exponential factor A is 10^5

73) **Explain Question:** We must identify which factor affects the rate constant (k) of a reaction.
Concept: This question is based on (factors affecting rete constant.
Solution:
Check each option:
1. **Concentration of reactant** → Does not affect k
   Concentration changes rate, not the rate constant.
2. **Temperature** → Affects k
   According to the Arrhenius equation: $K = Ae^{-E_a/RT,}$ k increases with temperature.
3. **Pressure** → Does not affect k directly
   Pressure affects rete only for gases by changing concentration, not the intrinsic rate constant.
4. **Concentration of product** → Does not affect k
   Product concentration affects equilibrium, not rate constant.

Final Answer: Option (2)

74)

Based on fact

75)

Based on fact

76) $E_a = \frac{2}{3}(E_{a_1} + E_{a_2} - E_{a_3})$

$= \frac{2}{3} \times (180 + 80 - 50)$

= 140 kJ/mol

77) $A \rightarrow B$

Initially $[A]_0$

t = t $[A]_0 - x$ nx

At intersection point

[A] = [B]

$[A]_0 - x = nx$

$\Rightarrow x = \frac{[A]_0}{n+1}$ & $[B] = \frac{n[A]_0}{n+1}$

78)

For Ist order rxn

$t\frac{1}{2} = \frac{\ln 2}{k}$

$\Rightarrow 6.93 \times 10^{-3} = \frac{0.693}{k}$

$\Rightarrow 6.93 \times 10^{-2} = \frac{0.693}{k}$

$\Rightarrow k = \frac{1}{10^{-2}}$

$\Rightarrow k = 10^2$

$\because \log 10^2 = 14 - \frac{1.2 \times 10^4}{T}$

$\Rightarrow 2 = 14 - 14 - \frac{1.2 \times 10^4}{T}$

$\Rightarrow -12 = -\frac{1.2 \times 10^4}{T}$

$\Rightarrow T = \frac{1.2 \times 10^3}{12}$

$\Rightarrow T = 1000k$

79) **Explanation:**

Reaction: $NH_3(g) \rightleftharpoons \frac{1}{2} N_2(g) + \frac{3}{2} H_2(g)$ $K_C = 4$

Reverse reaction: $N_2(g) + 3H_2(g) \rightleftharpoons 2\ NH_3(g)$ K_C'

Concept: Relation between K_p and K_p

Solution:

The relation between Kp and Kc is given by:

$K_C' = \frac{1}{(K_C)^2} = \frac{1}{(4)^2}$

$\square K_P = K_C(RT)^{\Delta n_g}$

$= \frac{1}{(4)^2}(RT)^{-2}$

$K_P = \left(\frac{1}{4 \times 800\ R}\right)^2$

Answer is option (3).

80) **Question Explanation:**
We are given a combustion equilibrium of methane and asked which statement is not true.
Concept:
• For equilibrium expressions, pure liquids (like $H_2O(\square)$) are excluded from Kp or Kc.
• The reaction is exothermic ($\Delta H < 0$).
• Adding reactants (CH_4 or O_2) shifts equilibrium to the right (Le Chatelier's principle).
• Concentrations of CO_2 and H_2O need not be equal.
Solution : Let's write equilibrium constant expression correctly :

$K_p = \frac{P_{CO_2}}{P_{CH_4}(P_{O_2})^2}$

But in the given option (2), they wrote

$K_p = \frac{[CO_2]}{[CH_4][O_2]}$ → missing square on $[O_2]$

Final Answer: option (2)

81)

Explain Question:
To find out the temp. at which $K_P = K_C$.

Concept:
Relation between K_P and K_C.

Solution:
$K_P = K_C\ (RT)^{\Delta ng}$
$\Rightarrow K_P = K_C$ if $(RT)^{\Delta ng} = 1$
If $\Delta ng = 0 \Rightarrow RT = 1$

$T = \frac{1}{R} = \frac{1}{0.0821}$

$= 12.18$ K

Final answer: Option (3)

82) **Explain the question :** We need to find direction of reaction if value of K_c and concentrations of reactants and product are given.
Concept : This question is based on reaction quotient.
Solution :
$A(g) + 2B(g) \rightleftharpoons 2C\ (g) + D\ (S)$

$$Q_c = \frac{[C]^2}{[A][B]^2}$$

$\because [C] = \frac{1}{10}, [A] = \frac{2}{10} [B] = \frac{3}{10}$

$$Q_C = \frac{\left(\frac{1}{10}\right)^2}{\left(\frac{2}{10}\right)\left(\frac{3}{10}\right)^3} = 0.55$$

– $Q_c < K_c$ $\therefore K_c = 3.6$

– reaction will proceed in forward direction.

Final answer : 1

83)

Explain the Question : We need to find how many times pressure of D change when pressure of B is doubled if reaction is given.

Concept : This question is based on K_P involving partial pressure.

Solution : $A(s) + B(g) \rightleftharpoons 2C(s) + 2D(g)$

$$K_P = \frac{(P_D)^2}{(P_B)}$$

After doubling $P'_B = 2P_B$, $P'_D = xP_D$

$$\frac{(P_D)^2}{(P_B)} = \frac{(xP_D)^2}{2P_B} \Rightarrow \frac{x^2}{2} = 1 \Rightarrow x = \sqrt{2}$$

– Pressure of D becomes $\sqrt{2}$ times the initial pressure.

Final answer : 3

84) **Question Explanation:** We need to determine which reaction shows the highest and lowest ratio of $\frac{K_p}{K_c}$ among the given equilibria at 298 K.

Concept: This question is based on (Relation between K_p and K_c)

Solution:

The relation between K_p and K_c is: $K_p = K_c(RT)^{\Delta n_g}$

Therefore,

$$\frac{K_p}{K_c} = (RT)^{\Delta n_g}$$

Calculate Δn_g for each reaction:

(a) $N_2O_4(g) \rightleftharpoons 2NO_2(g) : \Delta n_g = 2 - 1 = +1$

(b) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g) : \Delta n_g = 2 - 3 = -1$

(c) $2HI(g) \rightleftharpoons H_2(g) + I_2(g) : \Delta n_g = 2 - 2 = 0$

(d) $X(g) + Y(g) \rightleftharpoons 4Z(g) : \Delta n_g = 4 - 2 = +2$

Highest value corresponds to maximum $\Delta n_g = +2 \rightarrow$ reaction (d)

Lowest value corresponds to minimum $\Delta n_g = -1 \rightarrow$ reaction (b)

Final Answer: Option (1).

85) **Explain the question :-** We need to find value of K_C for equilibrium reaction if degree of dissociation of PCl_5 is given.

Concept : Degree of dissociation and K_C.

Solution :

$PCl_5 \rightleftharpoons PCl_3 + Cl_2$

t=0 1 0 0

t_{eq} 1 – x x x

$\%\alpha = 80\%$; $80 = \frac{x}{1} \times 100 \Rightarrow = 0.8$

$K_C = \frac{[PCl_3][Cl_2]}{[PCl_5]} = \frac{0.8 \times 0.8}{1-0.8} = 3.2$

$K_C = 3.2$

Final answer : 4

86) **Explain the question :** We need to find value of K_C for given reaction if initial moles of N_2 and H_2 and % α of N_2 are given. **Concept :** This question is based on degree of dissociation and K_C.

Solution :

$N_2(g) + 3H_2(g) \leftrightharpoons 2NH_3(g)$

t = 0 2 6 –

t_{eq} 2 – x 6 – 3x 2x

$(\%\alpha)N_2 = 50 \Rightarrow \frac{50}{100} = \frac{x}{2} \Rightarrow x = 1$

$K_C = \frac{[NH_3]^2}{[N_2][H_2]^3} = \frac{(2 \times 1)^2}{(2-1)(6-3 \times 1)^2}$

$K_C = \frac{2 \times 2}{3 \times 3 \times 3} = \frac{4}{27}$

Final answer = 1

87)

$3A_{2(g)} \rightleftharpoons 2A_{3(g)}$

t = 0 a 0

eq^n a–3x 2x x = aα

$\frac{2x}{3}$

a–x

$\frac{2a\alpha}{3}$

a–aα

$n_T = a - \frac{a\alpha}{3}$

88)

$A(g) + B(g) \rightleftharpoons C(g) + D(g)$ x = 0.5

t = 0 1 1 0 0

eq^m 1–x 1–x x x

0.5 0.5 0.5 0.5

$K_c = \frac{0.5 \times 0.5}{0.5 \times 0.5} = 1$

New eq^m

t = 0 1 2 0 0

(1–x) (2–x) x x

$$K_c = \frac{x \times x}{(1-x)(2-x)} = 1$$

$x^2 = 2–3x + x^2$

$x = 2/3$

$$\alpha_B = \frac{x}{2} \times 100$$

$$\alpha_B = \frac{2/3}{2} \times 100 = 33.33\ \%$$

89)

	N_2	+	$3H_2$	⇌	$2NH_3$
eq^m state	1.6		0.8		0.8

$$P_{NH_3} = P_T - (P_{N_2} + P_{H_2}) = 0.8$$

$$K_p = \frac{(0.8)^2}{(1.6)(0.8)^3} = 0.78$$

90) **Explain Question:**
Which of the following does not change concentration of NH_3 at equilibrium.
Concept:
Application of Le-chatelier principle.
Solution:
$N_2 + 3H_2 \rightleftharpoons 2NH_3\ \Delta H = –x$ KJ
(1) Increase of pressure, shift the reaction in direction of less moles ⇒ Forward Shift
⇒ This will increase concentration of NH_3
(2) Increase in temp. shift the exothermic reaction in backward direction. This will decrease concentration of NH_3.
(3) Decrease in volume. shift the reaction in direction of less moles. forward shift. This increase concentration of NH_3.
(4) Addition of catalyst does not shift equilibrium. So there is no change in concentration of reactant and product.
Final answer: Option (4)

BIOLOGY

91)

NCERT XII Pg # 117

92)

Question explanation: This question asks about the mutation rate of RNA and the evolutionary rate of RNA viruses.

Concept: Properties Of Genetic Material

Solution:

A. Statement I: False – RNA mutates faster, not slower, because RNA polymerases lack proofreading ability.

B. Statement II: True – RNA viruses with short life spans mutate and evolve faster due to frequent replication errors.

Final Answer: Statement I is false but Statement II is true.

93) NCERT Pg. # 168

94) NCERT-XII, Pg. # 88

95)

NCERT XII Pg # 115

96)

NCERT-XII, Pg#95,99

97) NCERT (XII) Pg # 93

98) **Question Explanation : RNA Pol functions in transcription**
Concept: Molecular Basis of Inheritance.
Solution:
In **eukaryotic cells**, different classes of RNA are synthesized by specific RNA polymerases:

A. **RNA Polymerase I (Pol I):** Transcribes ribosomal RNAs (rRNAs), specifically the major ones like 28S, 18S, and 5.8S rRNA.
B. **RNA Polymerase II (Pol II):** Transcribes the precursor of messenger RNA, which is **hnRNA** (heterogeneous nuclear RNA).
C. **RNA Polymerase III (Pol III):** Transcribes transfer RNAs (**t-RNA**) and the small ribosomal RNA (5S rRNA).

Prokaryotes (like bacteria) use a **single RNA Polymerase** (denoted generally as 'RNA Polymerase' in Column II, option i) for all major RNA types, including **16S rRNA** (a prokaryotic ribosomal subunit).
Final answer: Option (1)

99)

NCERT Pg. # 98,99

100)

NCERT-XII, Pg#95

101) NCERT Pg. # 97

102) NCERT-XII Pg. # 94,95

103)

NCERT XII Pg # 114

104)

Question Explanation: The question asks about the purpose of gel electrophoresis.

Concept: Gel electrophoresis

Solution: Gel electrophoresis is a technique that separates DNA (or RNA/protein) fragments based on size and charge.Smaller fragments move faster through the gel, larger fragments move slower.

Final answer: Option 4

105) NCERT (XII) Pg # 98, 99

106) NCERT-XII Pg. # 90

107)

NCERT-XII, Pg#91

108)

NCERT XII Pg # 108

109)

NCERT, Pg # 99

110) NCERT-XII, Pg. # 84

111) NCERT XII Page. No.168

112)

NCERT Pg. # 92

113) NCERT-XII Pg. # 94

114) NCERT XII Pg. No. # 114,115

115) NCERT (XIIth) Pg. # 94, 92, 91

116) NCERT XII Pg#104(E), 114(H)

117) NCERT XII Pg. 112

118) NCERT (XII) Pg. # 114,115

119)

Question Explanation: The question asks to identify incorrect statements about the genetic code.
Concept: Genetic Code
Solution:
Six codons actually code for amino acids like leucine, serine, and arginine, so statement (A) is incorrect.
Statements (B), (C), and (D) are correct: codons are read contiguously, three codons are stop codons, and AUG codes for methionine.

Final Answer: Option (4)

120) NCERT Pg. # 69

121) NCERT-XII Pg#101

122) NCERT XII Pg. No. # 106, 103

123) NCERT XII Pg. No. # 106

124) NCERT XII Pg. No. # 110

125) NCERT XII Pg. No. # 110, 112, 115

126) NCERT XII Pg. No. # 111

127) NCERT XII Pg. No. # 108, 112

128)

NCERT XII Pg # 101

129) NCERT XII Pg. No. # 112

130) NCERT Pg. No. # 104,111,109,114

131) NCERT XII Pg. No. # 117

132) NCERT XII Pg. No. 117

133) NCERT XII Pg. No. # 87, 113, 121

134) NCERT XII Pg. No. # 115

135) NCERT XII Pg. No. # 117,111

136)

NCERT-XII, Pg # 37

137) NCERT (XIIth) Pg. # 50, 51, Para-1, 1

138)

NCERT Pg. # 33

139)

NCERT Pg. # 34

140)

NCERT Pg. # 34

141) NCERT Pg.# 54

142)

NCERT Pg. # 34

143)

NCERT Pg. # 47

144) Module-1, Pg. # 38

145)

NCERT Pg. # 37

146)

NCERT Pg. # 34

147)

NCERT Pg. 50

148)

Generated by Allie

Problem Statement: The question asks to identify the correct sequence of developmental stages represented by the labels A, B, C, and D in a biological cycle, involving the union of sperm and ovum and subsequent stages leading to a foetus.

Underlying Concept: The problem is based on the stages of early embryonic development in humans, starting from fertilization to the formation of the foetus. The stages include zygote formation after fusion of sperm and ovum, cleavage resulting in the morula, development into blastocyst at implantation, and gastrulation where germ layers form.

Tips and Tricks: Remember the sequence: Zygote -> Morula -> Blastocyst -> Gastrula. Associating each stage with its location or event (e.g., implantation corresponds to blastocyst) helps in recall.

Common Mistakes: Confusing the order of stages, such as mistaking morula for blastocyst or interchanging gastrula and cleavage stages, or misunderstanding where implantation occurs.

Why Other Options Are Incorrect?: Option 1 incorrectly starts with 'Foetus' as A which is a much later stage. Option 3 misplaces 'Cleavage' as stage D rather than an ongoing process early in development. Option 4 incorrectly lists 'Gastrula' before 'Cleavage', disrupting the chronological order.

149) NCERT XII Pg. # 60 (E) 67 (H) Fig. 4.2

150) NCERT, Pg. # 32

151) NCERT, Pg. # 32

152) NCERT Pg # 37

153) NCERT Pg. # 37

154) NCERT Pg. # 37

155) NCERT Pg. # 38

156)

NCERT Pg. # 34

157)

Question Explanation: This question asks which hormone is responsible for milk formation in human females.

Concept: Mammary gland

Solution: Prolactin stimulates milk synthesis/formation in the mammary glands. Oxytocin is responsible for milk ejection (let-down reflex), not formation. Estrogen and progesterone help in mammary gland development during pregnancy.

Final answer: OPTION (1)

158)

NCERT Pg. # 36

159)

Explanation Question : to identify the hormone that attains two peaks during menstrual cycle.

Concept : This question is based on menstrual cycle.

Solution : Estrogen is the hormone that attains two peaks during menstrual cycle.

Final Answer : option (2) Estrogen

160)

NCERT Pg. 54

161)

The correct answer is **1. LH**.

A. **LH Surge:** Luteinizing hormone (LH) plays a crucial role in triggering ovulation.

   A. As the ovarian follicle matures, it produces increasing amounts of estrogen.
   B. High levels of estrogen provide a positive feedback loop, causing a surge in LH secretion from the pituitary gland.
   C. This LH surge causes the rupture of the mature Graafian follicle, releasing the ovum.

162)

Explain Question : to Match the Column–A with Column–B

Concept : Menstrual cycle

Explanation :

(A)Follicular phase → (ii) Development of follicle

(B) Luteal phase → (iii) Formation of corpus luteum

(C)Menstrual phase → (iv) Breakdown of endometrium

(D) Ovulatory phase → (i) Release of secondary oocyte

Final answer : 3

163)

NCERT Pg. # 44

164) Module-1 Pg # 46 Diagram

165)

NCERT Pg. # 44

166)

NCERT XII, Pg # 63

167) NCERT XII, Page # 60, 61

168)

NCERT Pg. # 44

169) NCERT Page # 64

170)

NCERT Pg. # 44

171) NCERT Pg # 62

172) NCERT Pg # 58

173)

NCERT-XII, Pg#60,62

174) NCERT–XII, Last para, Pg. # 57

175)

Question Explanation: Identify the incorrect disease–causative agent pair.

Concept: The question is based on STDs

Solution: The incorrect match is given in option 3.

A. Trichomoniasis – Trichomonas vaginalis - correct match
B. Genital herpes – Herpes simplex virus - correct match
C. **Gonorrhoea – Treponema pallidum - incorrect match** (Treponema pallidum causes syphilis; gonorrhoea is caused by Neisseria gonorrhoeae)
D. Genital warts – Human papilloma virus - correct match

Final answer: Option 3

176) NCERT Page No. 126

177) NCERT Page No. 127

178) NCERT Page No. 127

179) NCERT Page No. 127

180) NCERT Page No. 127